

Product Design Requirements (PDR)

FinBench `maths_lib` — Deterministic Financial Formula Library

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Parent System: FinBench Multi-Agent Business Analyst AI (PDR-BAAAI-001)
Total Formulas: 910 (target was 600; delivered 910)
Status: APPROVED FOR BUILD

1. Purpose & Rationale

`maths_lib` is the deterministic computation core of FinBench. It exists because large language models are unreliable at multi-step arithmetic and formula recall. Instead of asking Llama 3.1 8B to *remember* how to compute Black-Scholes or Macaulay Duration, the model only needs to *recognize* the formula and call a pre-built, unit-tested Python function that returns an exact answer.

This is the **"banana peel" strategy**: once the formula library is fully built (peeled), every reasoning question becomes easy to consume. The LLM becomes an orchestrator, not a calculator.

Design goals:

- **100% deterministic** — same inputs always yield same output (no LLM in the math path).
- **Zero cost** — pure Python + NumPy/SciPy. No paid APIs (satisfies C1).
- **Fully local** — no network calls (satisfies C2).
- **Fast** — every formula executes in under 10 milliseconds.
- **Auditable** — each result carries the formula name, inputs used, and the expression evaluated.
- **Reproducible** — `seed=42` wherever randomness appears, e.g. Monte Carlo (satisfies C5).

2. Domain Overview

The library is organized into **12 domains** totaling **910 unique formulas**. Each domain maps to exactly one Python module.

#	Domain	Module File	Formula Count
1	Profitability & Margin Ratios	<code>profitability.py</code>	50
2	Liquidity, Solvency & Efficiency Ratios	<code>liquidity_solvency.py</code>	50
3	Valuation Metrics & Models	<code>valuation.py</code>	60
4	Time-Series & Technical Analysis	<code>technical.py</code>	80
5	Options Pricing & Derivatives	<code>options.py</code>	45
6	Fixed Income & Bonds	<code>fixed_income.py</code>	50
7	Risk Management & Portfolio Theory	<code>risk.py</code>	50
8	Time Value of Money & Capital Budgeting	<code>tvm.py</code>	45
9	Corporate Finance & M&A	<code>corporate_ma.py</code>	50
10	Accounting & Depreciation	<code>accounting.py</code>	40
11	Statistics & Econometrics	<code>statistics_econ.py</code>	45
12	Core Math, Trigonometry, Linear Algebra & Geometry	<code>math_core.py</code>	45
13	Growth, Segment, Forensic & Modern Metrics	<code>growth_segment_forensic.py</code>	50
14	AI / Machine Learning Metrics & Functions	<code>ai_ml.py</code>	93
15	Probability Theory & Distributions	<code>probability.py</code>	55
16	Advanced Statistics & Hypothesis Testing	<code>statistics_advanced.py</code>	53
17	Business Analyst KPIs, Forecasting & Decision	<code>business_analyst.py</code>	49
	TOTAL		910

3. Library Structure (Directory Layout)

```
src/
└─ maths_lib/
   ├── __init__.py      # exports FORMULA_REGISTRY + all functions
   ├── registry.py      # name -> function map, metadata, lookup API
   ├── base.py          # FormulaResult dataclass, @formula decorator,
   │                   # input validation, safe-divide helpers
   ├── exceptions.py    # MissingInputError, DivisionGuard, DomainError
   ├── profitability.py # D01 - 50 formulas
   ├── liquidity_solveny.py # D02 - 50 formulas
   ├── valuation.py     # D03 - 60 formulas
   ├── technical.py     # D04 - 80 formulas
   ├── options.py       # D05 - 45 formulas
   ├── fixed_income.py  # D06 - 50 formulas
   ├── risk.py          # D07 - 50 formulas
   ├── tvml.py          # D08 - 45 formulas
   ├── corporate_ma.py  # D09 - 50 formulas
   ├── accounting.py    # D10 - 40 formulas
   ├── statistics_econ.py # D11 - 45 formulas
   └─ math_core.py      # D12 - 45 formulas

tests/
└─ maths_lib/
   ├── test_profitability.py # >=3 cases per formula
   ├── test_liquidity_solveny.py
   ├── test_valuation.py
   ├── test_technical.py
   ├── test_options.py
   ├── test_fixed_income.py
   ├── test_risk.py
   ├── test_tvm.py
   ├── test_corporate_ma.py
   ├── test_accounting.py
   ├── test_statistics_econ.py
   ├── test_math_core.py
   ├── test_registry.py     # registry integrity, unique names
   └─ golden/               # known-answer reference values (JSON)
```

4. Core Contracts (base.py)

Every formula function follows one signature pattern and returns a `FormulaResult`. This is what makes the library auditable and lets the Composite Resolver (N20) chain formulas together.

```

from dataclasses import dataclass, field
from typing import Callable, Dict, List, Optional
import functools

@dataclass
class FormulaResult:
    value: float # the computed answer
    formula_id: str # e.g. 'gross_margin'
    formula_name: str # e.g. 'Gross Margin %'
    expression: str # e.g. '(Revenue - COGS) / Revenue * 100'
    inputs_used: Dict[str, float] # {'revenue': 383285, 'cogs': 214137}
    unit: str = '' # '%', 'x', '$', 'days', ''
    domain: str = '' # 'D01_profitability'
    valid: bool = True
    error: Optional[str] = None

FORMULA_REGISTRY: Dict[str, dict] = {}

def formula(fid, name, expression, domain, unit=''):
    """Decorator: registers a function and attaches metadata."""
    def wrap(fn):
        FORMULA_REGISTRY[fid] = {
            'fn': fn, 'name': name, 'expression': expression,
            'domain': domain, 'unit': unit,
            'inputs': list(fn.__code__.co_varnames[:fn.__code__.co_argcount]),
        }
        @functools.wraps(fn)
        def inner(*a, **k):
            return fn(*a, **k)
        return inner
    return wrap

def safe_div(numerator, denominator):
    """Division guard - returns None instead of raising ZeroDivisionError."""
    if denominator == 0 or denominator is None:
        return None
    return numerator / denominator

```

Example formula implementation:

```

@formula('gross_margin', 'Gross Margin %',
        '(Revenue - COGS) / Revenue * 100', 'D01_profitability', unit='%')
def gross_margin(revenue: float, cogs: float) -> FormulaResult:
    val = safe_div((revenue - cogs), revenue)
    return FormulaResult(
        value=(val * 100) if val is not None else None,
        formula_id='gross_margin', formula_name='Gross Margin %',
        expression='(Revenue - COGS) / Revenue * 100',
        inputs_used={'revenue': revenue, 'cogs': cogs},
        unit='%', domain='D01_profitability',
        valid=(val is not None),
        error=None if val is not None else 'division by zero (revenue=0)',
    )

```

5. Complete Formula Catalog

All 910 formulas enumerated by domain. Columns: # (sequence), ID (function name), Name, Description, Formula (expression), Inputs.

5.1 Profitability & Margin Ratios · [profitability.py](#) · 50 formulas

#	ID	Name	Description	Formula	Inputs
1	gross_margin	Gross Margin %	Profit left after COGS as % of revenue	$(\text{Revenue} - \text{COGS}) / \text{Revenue} * 100$	revenue, cogs
2	gross_profit	Gross Profit	Revenue minus cost of goods sold	$\text{Revenue} - \text{COGS}$	revenue, cogs
3	operating_margin	Operating Margin %	Operating income as % of revenue	$\text{Operating_Income} / \text{Revenue} * 100$	operating_income, revenue
4	net_margin	Net Profit Margin %	Net income as % of revenue	$\text{Net_Income} / \text{Revenue} * 100$	net_income, revenue
5	ebitda_margin	EBITDA Margin %	EBITDA as % of revenue	$\text{EBITDA} / \text{Revenue} * 100$	ebitda, revenue
6	ebit_margin	EBIT Margin %	EBIT as % of revenue	$\text{EBIT} / \text{Revenue} * 100$	ebit, revenue

#	ID	Name	Description	Formula	Inputs
7	pretax_margin	Pretax Margin %	Pretax income as % of revenue	$\text{Pretax_Income} / \text{Revenue} * 100$	pretax_income, revenue
8	contribution_margin	Contribution Margin	Revenue minus variable costs	$\text{Revenue} - \text{Variable_Costs}$	revenue, variable_costs
9	contribution_margin_ratio	Contribution Margin Ratio %	Contribution margin as % of revenue	$(\text{Revenue} - \text{Variable_Costs}) / \text{Revenue} * 100$	revenue, variable_costs
10	fcf_margin	Free Cash Flow Margin %	FCF as % of revenue	$\text{FCF} / \text{Revenue} * 100$	fcf, revenue
11	ocf_margin	Operating Cash Flow Margin %	OCF as % of revenue	$\text{OCF} / \text{Revenue} * 100$	ocf, revenue
12	return_on_equity	Return on Equity (ROE) %	Net income as % of shareholders equity	$\text{Net_Income} / \text{Shareholders_Equity} * 100$	net_income, shareholders_equity
13	return_on_assets	Return on Assets (ROA) %	Net income as % of total assets	$\text{Net_Income} / \text{Total_Assets} * 100$	net_income, total_assets
14	return_on_invested_capital	Return on Invested Capital (ROIC) %	NOPAT as % of invested capital	$\text{NOPAT} / \text{Invested_Capital} * 100$	nopat, invested_capital
15	return_on_capital_employed	Return on Capital Employed (ROCE) %	EBIT as % of capital employed	$\text{EBIT} / \text{Capital_Employed} * 100$	ebit, capital_employed
16	return_on_sales	Return on Sales (ROS) %	Operating profit as % of sales	$\text{Operating_Income} / \text{Sales} * 100$	operating_income, sales
17	return_on_tangible_equity	Return on Tangible Equity %	Net income / tangible equity	$\text{Net_Income} / (\text{Equity} - \text{Intangibles}) * 100$	net_income, equity, intangibles
18	return_on_net_assets	Return on Net Assets (RONA) %	Net income / fixed assets + working capital	$\text{Net_Income} / (\text{Fixed_Assets} + \text{Working_Capital}) * 100$	net_income, fixed_assets, working_capital
19	nopat	Net Operating Profit After Tax	Operating income after tax	$\text{EBIT} * (1 - \text{Tax_Rate})$	ebit, tax_rate
20	ebitda	EBITDA	Earnings before interest, tax, depreciation, amortization	$\text{Net_Income} + \text{Interest} + \text{Taxes} + \text{Depreciation} + \text{Amortization}$	net_income, interest, taxes, depreciation, amortization
21	ebit	EBIT	Earnings before interest and tax	$\text{Net_Income} + \text{Interest} + \text{Taxes}$	net_income, interest, taxes
22	effective_tax_rate	Effective Tax Rate %	Tax expense as % of pretax income	$\text{Tax_Expense} / \text{Pretax_Income} * 100$	tax_expense, pretax_income
23	operating_leverage	Degree of Operating Leverage	% change EBIT / % change sales	$\text{Pct_Change_EBIT} / \text{Pct_Change_Sales}$	pct_change_ebit, pct_change_sales
24	financial_leverage	Degree of Financial Leverage	% change EPS / % change EBIT	$\text{Pct_Change_EPS} / \text{Pct_Change_EBIT}$	pct_change_eps, pct_change_ebit
25	combined_leverage	Degree of Combined Leverage	DOL times DFL	$\text{DOL} * \text{DFL}$	dol, dfl
26	dupont_roe_3step	DuPont ROE (3-Step)	Net margin x asset turnover x equity multiplier	$\text{Net_Margin} * \text{Asset_Turnover} * \text{Equity_Multiplier}$	net_margin, asset_turnover, equity_multiplier
27	dupont_roe_5step	DuPont ROE (5-Step)	Extended DuPont with tax + interest burden	$\text{Tax_Burden} * \text{Interest_Burden} * \text{Operating_Margin} * \text{Asset_Turnover} * \text{Equity_Multiplier}$	tax_burden, interest_burden, operating_margin, asset_turnover, equity_multiplier
28	tax_burden	Tax Burden Ratio	Net income / pretax income	$\text{Net_Income} / \text{Pretax_Income}$	net_income, pretax_income
29	interest_burden	Interest Burden Ratio	Pretax income / EBIT	$\text{Pretax_Income} / \text{EBIT}$	pretax_income, ebit
30	equity_multiplier	Equity Multiplier	Total assets / equity	$\text{Total_Assets} / \text{Shareholders_Equity}$	total_assets, shareholders_equity
31	operating_ratio	Operating Ratio %	Operating costs / revenue	$\text{Operating_Costs} / \text{Revenue} * 100$	operating_costs, revenue
32	cost_of_revenue_ratio	Cost of Revenue Ratio %	COGS / revenue	$\text{COGS} / \text{Revenue} * 100$	cogs, revenue

#	ID	Name	Description	Formula	Inputs
33	overhead_ratio	Overhead Ratio %	Operating expenses / (net interest + operating income)	$\text{Operating_Expenses} / (\text{Net_Interest} + \text{Operating_Income}) * 100$	operating_expenses, net_interest, operating_income
34	sga_ratio	SG&A to Revenue %	Selling general admin / revenue	$\text{SGA} / \text{Revenue} * 100$	sga, revenue
35	rnd_ratio	R&D to Revenue %	Research dev / revenue	$\text{RnD} / \text{Revenue} * 100$	rnd, revenue
36	rnd_intensity	R&D Intensity %	R&D spend / revenue	$\text{RnD_Expense} / \text{Revenue} * 100$	rnd_expense, revenue
37	net_income_growth	Net Income Growth %	YoY net income change	$(\text{NI_Current} - \text{NI_Prior}) / \text{NI_Prior} * 100$	ni_current, ni_prior
38	revenue_growth	Revenue Growth %	YoY revenue change	$(\text{Rev_Current} - \text{Rev_Prior}) / \text{Rev_Prior} * 100$	rev_current, rev_prior
39	operating_income_growth	Operating Income Growth %	YoY operating income change	$(\text{OI_Current} - \text{OI_Prior}) / \text{OI_Prior} * 100$	oi_current, oi_prior
40	eps_basic	Basic EPS	Net income / weighted basic shares	$(\text{Net_Income} - \text{Pref_Dividends}) / \text{Basic_Shares}$	net_income, pref_dividends, basic_shares
41	eps_diluted	Diluted EPS	Net income / diluted shares	$(\text{Net_Income} - \text{Pref_Dividends}) / \text{Diluted_Shares}$	net_income, pref_dividends, diluted_shares
42	eps_growth	EPS Growth %	YoY EPS change	$(\text{EPS_Current} - \text{EPS_Prior}) / \text{EPS_Prior} * 100$	eps_current, eps_prior
43	cash_return_on_assets	Cash Return on Assets %	OCF / total assets	$\text{OCF} / \text{Total_Assets} * 100$	ocf, total_assets
44	cash_roe	Cash Return on Equity %	OCF / equity	$\text{OCF} / \text{Shareholders_Equity} * 100$	ocf, shareholders_equity
45	gross_profit_growth	Gross Profit Growth %	YoY gross profit change	$(\text{GP_Current} - \text{GP_Prior}) / \text{GP_Prior} * 100$	gp_current, gp_prior
46	ebitda_growth	EBITDA Growth %	YoY EBITDA change	$(\text{EBITDA_Current} - \text{EBITDA_Prior}) / \text{EBITDA_Prior} * 100$	ebitda_current, ebitda_prior
47	incremental_margin	Incremental Margin %	Change in profit / change in revenue	$\text{Delta_Profit} / \text{Delta_Revenue} * 100$	delta_profit, delta_revenue
48	breakeven_point_units	Breakeven Point (Units)	Fixed costs / contribution per unit	$\text{Fixed_Costs} / (\text{Price} - \text{Variable_Cost_Per_Unit})$	fixed_costs, price, variable_cost_per_unit
49	breakeven_point_revenue	Breakeven Point (Revenue)	Fixed costs / contribution margin ratio	$\text{Fixed_Costs} / \text{Contribution_Margin_Ratio}$	fixed_costs, contribution_margin_ratio
50	margin_of_safety	Margin of Safety %	(Sales - breakeven) / sales	$(\text{Sales} - \text{Breakeven_Sales}) / \text{Sales} * 100$	sales, breakeven_sales

5.2 Liquidity, Solvency & Efficiency Ratios · [liquidity_solvency.py](#) · 50 formulas

#	ID	Name	Description	Formula	Inputs
51	current_ratio	Current Ratio	Current assets / current liabilities	$\text{Current_Assets} / \text{Current_Liabilities}$	current_assets, current_liabilities
52	quick_ratio	Quick Ratio (Acid Test)	Liquid assets / current liabilities	$(\text{Current_Assets} - \text{Inventory}) / \text{Current_Liabilities}$	current_assets, inventory, current_liabilities
53	cash_ratio	Cash Ratio	Cash + equivalents / current liabilities	$(\text{Cash} + \text{Marketable_Securities}) / \text{Current_Liabilities}$	cash, marketable_securities, current_liabilities
54	operating_cash_flow_ratio	Operating Cash Flow Ratio	OCF / current liabilities	$\text{OCF} / \text{Current_Liabilities}$	ocf, current_liabilities
55	working_capital	Working Capital	Current assets minus current liabilities	$\text{Current_Assets} - \text{Current_Liabilities}$	current_assets, current_liabilities
56	working_capital_ratio	Working Capital Ratio	Same as current ratio	$\text{Current_Assets} / \text{Current_Liabilities}$	current_assets, current_liabilities
57	net_working_capital_to_sales	NWC to Sales %	Net working capital / sales	$\text{Working_Capital} / \text{Sales} * 100$	working_capital, sales
58	defensive_interval_ratio	Defensive Interval Ratio (days)	Liquid assets / daily expenses	$\text{Liquid_Assets} / \text{Daily_Operating_Expenses}$	liquid_assets, daily_operating_expenses
59	debt_to_equity	Debt-to-Equity	Total debt / equity	$\text{Total_Debt} / \text{Shareholders_Equity}$	total_debt, shareholders_equity

#	ID	Name	Description	Formula	Inputs
60	debt_to_assets	Debt-to-Assets	Total debt / total assets	$\text{Total_Debt} / \text{Total_Assets}$	total_debt, total_assets
61	debt_to_capital	Debt-to-Capital	Debt / (debt + equity)	$\text{Total_Debt} / (\text{Total_Debt} + \text{Shareholders_Equity})$	total_debt, shareholders_equity
62	debt_to_ebitda	Debt-to-EBITDA	Total debt / EBITDA	$\text{Total_Debt} / \text{EBITDA}$	total_debt, ebitda
63	net_debt	Net Debt	Total debt minus cash	$\text{Total_Debt} - \text{Cash}$	total_debt, cash
64	net_debt_to_ebitda	Net Debt-to-EBITDA	Net debt / EBITDA	$(\text{Total_Debt} - \text{Cash}) / \text{EBITDA}$	total_debt, cash, ebitda
65	equity_ratio	Equity Ratio	Equity / total assets	$\text{Shareholders_Equity} / \text{Total_Assets}$	shareholders_equity, total_assets
66	financial_leverage_ratio	Financial Leverage Ratio	Total assets / equity	$\text{Total_Assets} / \text{Shareholders_Equity}$	total_assets, shareholders_equity
67	interest_coverage	Interest Coverage Ratio	EBIT / interest expense	$\text{EBIT} / \text{Interest_Expense}$	ebit, interest_expense
68	ebitda_coverage	EBITDA Coverage Ratio	EBITDA / interest expense	$\text{EBITDA} / \text{Interest_Expense}$	ebitda, interest_expense
69	fixed_charge_coverage	Fixed Charge Coverage	(EBIT + lease) / (interest + lease)	$(\text{EBIT} + \text{Lease_Payments}) / (\text{Interest_Expense} + \text{Lease_Payments})$	ebit, lease_payments, interest_expense
70	times_interest_earned	Times Interest Earned	EBIT / interest expense	$\text{EBIT} / \text{Interest_Expense}$	ebit, interest_expense
71	debt_service_coverage	Debt Service Coverage Ratio (DSCR)	Net operating income / debt service	$\text{Net_Operating_Income} / \text{Total_Debt_Service}$	net_operating_income, total_debt_service
72	cash_flow_to_debt	Cash Flow to Debt	OCF / total debt	$\text{OCF} / \text{Total_Debt}$	ocf, total_debt
73	capitalization_ratio	Capitalization Ratio	LT debt / (LT debt + equity)	$\text{Long_Term_Debt} / (\text{Long_Term_Debt} + \text{Shareholders_Equity})$	long_term_debt, shareholders_equity
74	asset_turnover	Asset Turnover	Revenue / total assets	$\text{Revenue} / \text{Total_Assets}$	revenue, total_assets
75	fixed_asset_turnover	Fixed Asset Turnover	Revenue / net PP&E	$\text{Revenue} / \text{Net_PPE}$	revenue, net_ppe
76	inventory_turnover	Inventory Turnover	COGS / average inventory	$\text{COGS} / \text{Average_Inventory}$	cogs, average_inventory
77	receivables_turnover	Receivables Turnover	Revenue / average receivables	$\text{Revenue} / \text{Average_Receivables}$	revenue, average_receivables
78	payables_turnover	Payables Turnover	COGS / average payables	$\text{COGS} / \text{Average_Payables}$	cogs, average_payables
79	working_capital_turnover	Working Capital Turnover	Revenue / working capital	$\text{Revenue} / \text{Working_Capital}$	revenue, working_capital
80	equity_turnover	Equity Turnover	Revenue / equity	$\text{Revenue} / \text{Shareholders_Equity}$	revenue, shareholders_equity
81	total_capital_turnover	Total Capital Turnover	Revenue / total capital	$\text{Revenue} / \text{Total_Capital}$	revenue, total_capital
82	days_sales_outstanding	Days Sales Outstanding (DSO)	365 / receivables turnover	$365 / \text{Receivables_Turnover}$	receivables_turnover
83	days_inventory_outstanding	Days Inventory Outstanding (DIO)	365 / inventory turnover	$365 / \text{Inventory_Turnover}$	inventory_turnover
84	days_payable_outstanding	Days Payable Outstanding (DPO)	365 / payables turnover	$365 / \text{Payables_Turnover}$	payables_turnover
85	cash_conversion_cycle	Cash Conversion Cycle (days)	DSO + DIO - DPO	$\text{DSO} + \text{DIO} - \text{DPO}$	dso, dio, dpo
86	operating_cycle	Operating Cycle (days)	DSO + DIO	$\text{DSO} + \text{DIO}$	dso, dio
87	dso_direct	DSO Direct	Receivables / revenue x 365	$\text{Receivables} / \text{Revenue} * 365$	receivables, revenue
88	dio_direct	DIO Direct	Inventory / COGS x 365	$\text{Inventory} / \text{COGS} * 365$	inventory, cogs
89	dpo_direct	DPO Direct	Payables / COGS x 365	$\text{Payables} / \text{COGS} * 365$	payables, cogs
90	capital_intensity	Capital Intensity %	CapEx / revenue	$\text{CapEx} / \text{Revenue} * 100$	capex, revenue
91	capital_intensity_assets	Capital Intensity (Assets)	Total assets / revenue	$\text{Total_Assets} / \text{Revenue}$	total_assets, revenue

#	ID	Name	Description	Formula	Inputs
92	fixed_assets_to_equity	Fixed Assets to Equity	Net PP&E / equity	$\text{Net_PPE} / \text{Shareholders_Equity}$	net_ppe, shareholders_equity
93	long_term_debt_to_equity	LT Debt to Equity	LT debt / equity	$\text{Long_Term_Debt} / \text{Shareholders_Equity}$	long_term_debt, shareholders_equity
94	short_term_debt_ratio	Short-Term Debt Ratio	ST debt / total debt	$\text{Short_Term_Debt} / \text{Total_Debt}$	short_term_debt, total_debt
95	current_liabilities_ratio	Current Liabilities Ratio	Current liabilities / total liabilities	$\text{Current_Liabilities} / \text{Total_Liabilities}$	current_liabilities, total_liabilities
96	solvency_ratio	Solvency Ratio	(Net income + depreciation) / total liabilities	$(\text{Net_Income} + \text{Depreciation}) / \text{Total_Liabilities}$	net_income, depreciation, total_liabilities
97	financial_autonomy_ratio	Financial Autonomy Ratio	Equity / total liabilities	$\text{Shareholders_Equity} / \text{Total_Liabilities}$	shareholders_equity, total_liabilities
98	net_gearing	Net Gearing %	Net debt / equity	$(\text{Total_Debt} - \text{Cash}) / \text{Shareholders_Equity} * 100$	total_debt, cash, shareholders_equity
99	altman_z_score	Altman Z-Score	Bankruptcy predictor composite	$1.2*A + 1.4*B + 3.3*C + 0.6*D + 1.0*E$	a, b, c, d, e
100	piotroski_f_score	Piotroski F-Score	9-point financial strength score	Sum of 9 binary signals	signals_list

5.3 Valuation Metrics & Models · valuation.py · 60 formulas

#	ID	Name	Description	Formula	Inputs
101	pe_ratio	Price-to-Earnings (P/E)	Price / EPS	$\text{Price} / \text{EPS}$	price, eps
102	forward_pe	Forward P/E	Price / forward EPS	$\text{Price} / \text{Forward_EPS}$	price, forward_eps
103	peg_ratio	PEG Ratio	P/E / earnings growth	$\text{PE_Ratio} / \text{Earnings_Growth_Rate}$	pe_ratio, earnings_growth_rate
104	pb_ratio	Price-to-Book (P/B)	Price / book value per share	$\text{Price} / \text{Book_Value_Per_Share}$	price, book_value_per_share
105	ps_ratio	Price-to-Sales (P/S)	Market cap / revenue	$\text{Market_Cap} / \text{Revenue}$	market_cap, revenue
106	pcf_ratio	Price-to-Cash-Flow	Price / cash flow per share	$\text{Price} / \text{Cash_Flow_Per_Share}$	price, cash_flow_per_share
107	p_fcf_ratio	Price-to-Free-Cash-Flow	Market cap / FCF	$\text{Market_Cap} / \text{FCF}$	market_cap, fcf
108	ev	Enterprise Value	Market cap + debt - cash	$\text{Market_Cap} + \text{Total_Debt} - \text{Cash}$	market_cap, total_debt, cash
109	ev_ebitda	EV/EBITDA	Enterprise value / EBITDA	$\text{EV} / \text{EBITDA}$	ev, ebitda
110	ev_ebit	EV/EBIT	Enterprise value / EBIT	EV / EBIT	ev, ebit
111	ev_sales	EV/Sales	Enterprise value / revenue	$\text{EV} / \text{Revenue}$	ev, revenue
112	ev_fcf	EV/FCF	Enterprise value / FCF	EV / FCF	ev, fcf
113	dividend_yield	Dividend Yield %	Annual dividend / price	$\text{Annual_Dividend} / \text{Price} * 100$	annual_dividend, price
114	dividend_payout_ratio	Dividend Payout Ratio %	Dividends / net income	$\text{Dividends} / \text{Net_Income} * 100$	dividends, net_income
115	retention_ratio	Retention Ratio %	1 - payout ratio	$(1 - \text{Payout_Ratio}) * 100$	payout_ratio
116	book_value_per_share	Book Value Per Share	Equity / shares outstanding	$\text{Shareholders_Equity} / \text{Shares_Outstanding}$	shareholders_equity, shares_outstanding
117	tangible_book_value	Tangible Book Value	Equity - intangibles - goodwill	$\text{Shareholders_Equity} - \text{Intangibles} - \text{Goodwill}$	shareholders_equity, intangibles, goodwill
118	tangible_book_per_share	Tangible Book Value Per Share	Tangible BV / shares	$\text{Tangible_Book_Value} / \text{Shares_Outstanding}$	tangible_book_value, shares_outstanding
119	market_cap	Market Capitalization	Price x shares outstanding	$\text{Price} * \text{Shares_Outstanding}$	price, shares_outstanding
120	earnings_yield	Earnings Yield %	EPS / price	$\text{EPS} / \text{Price} * 100$	eps, price
121	fcf_yield	Free Cash Flow Yield %	FCF per share / price	$\text{FCF_Per_Share} / \text{Price} * 100$	fcf_per_share, price

#	ID	Name	Description	Formula	Inputs
122	dcf_value	DCF Present Value	Sum of discounted cash flows	$\text{Sum}(\text{CF}_t / (1+r)^t)$	cash_flows, discount_rate
123	dcf_two_stage	Two-Stage DCF	Explicit forecast + terminal value	$\text{Sum}(\text{CF}_t / (1+r)^t) + \text{TV} / (1+r)^n$	cash_flows, discount_rate, terminal_value
124	terminal_value_gordon	Terminal Value (Gordon)	$\text{FCF} \times (1+g) / (r-g)$	$\text{FCF} * (1 + g) / (r - g)$	fcf, growth_rate, discount_rate
125	terminal_value_exit	Terminal Value (Exit Multiple)	Final year metric x exit multiple	$\text{Final_EBITDA} * \text{Exit_Multiple}$	final_ebitda, exit_multiple
126	gordon_growth_model	Gordon Growth Model (DDM)	$D1 / (r - g)$	$D1 / (\text{Required_Return} - \text{Growth_Rate})$	d1, required_return, growth_rate
127	ddm_multistage	Multi-Stage DDM	Variable growth dividend model	$\text{Sum}(D_t / (1+r)^t) + \text{TV}$	dividends, discount_rate, terminal_value
128	fcff	Free Cash Flow to Firm	$\text{EBIT}(1-t) + \text{D\&A} - \text{CapEx} - \text{WC change}$	$\text{EBIT} * (1 - \text{Tax}) + \text{DA} - \text{CapEx} - \text{Delta_WC}$	ebit, tax_rate, da, capex, delta_wc
129	fcfe	Free Cash Flow to Equity	$\text{FCFF} - \text{interest}(1-t) + \text{net borrowing}$	$\text{FCFF} - \text{Interest} * (1 - \text{Tax}) + \text{Net_Borrowing}$	fcff, interest, tax_rate, net_borrowing
130	fcf_simple	Free Cash Flow (Simple)	$\text{OCF} - \text{CapEx}$	$\text{OCF} - \text{CapEx}$	ocf, capex
131	wacc	Weighted Average Cost of Capital	Weighted equity + debt cost	$\text{We} * \text{Re} + \text{Wd} * \text{Rd} * (1 - \text{Tax})$	weight_equity, cost_equity, weight_debt, cost_debt, tax_rate
132	cost_of_equity_capm	Cost of Equity (CAPM)	$\text{Rf} + \text{beta} \times (\text{Rm} - \text{Rf})$	$\text{Rf} + \text{Beta} * (\text{Rm} - \text{Rf})$	risk_free, beta, market_return
133	cost_of_equity_ddm	Cost of Equity (DDM)	$D1/P + g$	$D1 / \text{Price} + \text{Growth_Rate}$	d1, price, growth_rate
134	cost_of_debt	Cost of Debt (After-Tax)	Interest rate x (1-tax)	$\text{Interest_Rate} * (1 - \text{Tax_Rate})$	interest_rate, tax_rate
135	capm	CAPM Expected Return	$\text{Rf} + \text{beta} \times \text{equity premium}$	$\text{Rf} + \text{Beta} * (\text{Rm} - \text{Rf})$	risk_free, beta, market_return
136	fama_french_3	Fama-French 3-Factor	$\text{Rf} + \text{market} + \text{SMB} + \text{HML}$	$\text{Rf} + b1 * \text{MKT} + b2 * \text{SMB} + b3 * \text{HML}$	risk_free, b1, mkt, b2, smb, b3, hml
137	fama_french_5	Fama-French 5-Factor	$\text{FF3} + \text{RMW} + \text{CMA}$	$\text{Rf} + b1 * \text{MKT} + b2 * \text{SMB} + b3 * \text{HML} + b4 * \text{RMW} + b5 * \text{CMA}$	risk_free, betas, factors
138	residual_income	Residual Income	Net income - equity charge	$\text{Net_Income} - (\text{Equity} * \text{Cost_of_Equity})$	net_income, equity, cost_of_equity
139	eva	Economic Value Added	$\text{NOPAT} - (\text{capital} \times \text{WACC})$	$\text{NOPAT} - (\text{Invested_Capital} * \text{WACC})$	nopat, invested_capital, wacc
140	mva	Market Value Added	Market value - invested capital	$\text{Market_Value} - \text{Invested_Capital}$	market_value, invested_capital
141	justified_pe	Justified P/E	$\text{Payout} \times (1+g) / (r-g)$	$\text{Payout} * (1+g) / (r-g)$	payout, growth_rate, required_return
142	justified_pb	Justified P/B	$(\text{ROE} - g) / (r - g)$	$(\text{ROE} - g) / (r - g)$	roe, growth_rate, required_return
143	graham_number	Graham Number	$\text{sqrt}(22.5 \times \text{EPS} \times \text{BVPS})$	$\text{sqrt}(22.5 * \text{EPS} * \text{BVPS})$	eps, bvps
144	sum_of_parts	Sum-of-the-Parts Value	Sum of segment values	$\text{Sum}(\text{Segment_Value}_i)$	segment_values
145	net_asset_value	Net Asset Value (NAV)	Assets - liabilities	$\text{Total_Assets} - \text{Total_Liabilities}$	total_assets, total_liabilities
146	liquidation_value	Liquidation Value	Asset recovery - liabilities	$\text{Asset_Recovery_Value} - \text{Total_Liabilities}$	asset_recovery_value, total_liabilities
147	replacement_value	Replacement Value	Cost to rebuild assets	$\text{Replacement_Cost_Assets} - \text{Liabilities}$	replacement_cost_assets, liabilities
148	price_to_tangible_book	Price-to-Tangible-Book	Price / tangible BVPS	$\text{Price} / \text{Tangible_BVPS}$	price, tangible_bvps
149	ev_to_invested_capital	EV/Invested Capital	Enterprise value / invested capital	$\text{EV} / \text{Invested_Capital}$	ev, invested_capital
150	dividend_per_share	Dividend Per Share	Total dividends / shares	$\text{Total_Dividends} / \text{Shares_Outstanding}$	total_dividends, shares_outstanding
151	dividend_coverage	Dividend Coverage Ratio	EPS / DPS	$\text{EPS} / \text{Dividend_Per_Share}$	eps, dividend_per_share
152	total_shareholder_return	Total Shareholder Return %	$(\text{Price change} + \text{dividends}) / \text{start price}$	$(\text{Price_End} - \text{Price_Start} + \text{Dividends}) / \text{Price_Start} * 100$	price_end, price_start, dividends

#	ID	Name	Description	Formula	Inputs
153	<code>implied_growth_rate</code>	Implied Growth Rate	$r - D1/P$	<code>Required_Return - D1/Price</code>	<code>required_return</code> , <code>d1</code> , <code>price</code>
154	<code>ev_per_share</code>	EV Per Share	Enterprise value / shares	<code>EV / Shares_Outstanding</code>	<code>ev</code> , <code>shares_outstanding</code>
155	<code>price_to_nav</code>	Price-to-NAV	Price / NAV per share	<code>Price / NAV_Per_Share</code>	<code>price</code> , <code>nav_per_share</code>
156	<code>cape_ratio</code>	CAPE (Shiller P/E)	Price / 10yr avg real earnings	<code>Price / Avg_10yr_Real_EPS</code>	<code>price</code> , <code>avg_10yr_real_eps</code>
157	<code>rule_of_40</code>	Rule of 40 %	Revenue growth + profit margin	<code>Revenue_Growth_Pct + Profit_Margin_Pct</code>	<code>revenue_growth_pct</code> , <code>profit_margin_pct</code>
158	<code>magic_formula_yield</code>	Magic Formula Earnings Yield	EBIT / EV	<code>EBIT / EV</code>	<code>ebit</code> , <code>ev</code>
159	<code>owners_earnings</code>	Owner's Earnings (Buffett)	NI + D&A - maintenance CapEx	<code>Net_Income + DA - Maintenance_CapEx</code>	<code>net_income</code> , <code>da</code> , <code>maintenance_capex</code>
160	<code>intrinsic_value_growth</code>	Intrinsic Value (Growth)	EPS x (8.5 + 2g) Graham formula	<code>EPS * (8.5 + 2 * Growth_Rate)</code>	<code>eps</code> , <code>growth_rate</code>

5.4 Time-Series & Technical Analysis · `technical.py` · 80 formulas

#	ID	Name	Description	Formula	Inputs
161	<code>sma</code>	Simple Moving Average	Mean of last n prices	<code>Sum(Prices[-n:]) / n</code>	<code>prices</code> , <code>period</code>
162	<code>ema</code>	Exponential Moving Average	Weighted MA with decay	<code>Price*k + EMA_prev*(1-k)</code> , $k=2/(n+1)$	<code>prices</code> , <code>period</code>
163	<code>wma</code>	Weighted Moving Average	Linearly weighted MA	<code>Sum(Price_i * Weight_i) / Sum(Weights)</code>	<code>prices</code> , <code>period</code>
164	<code>dema</code>	Double EMA	$2*EMA - EMA(EMA)$	<code>2*EMA - EMA(EMA)</code>	<code>prices</code> , <code>period</code>
165	<code>tema</code>	Triple EMA	$3*EMA - 3*EMA2 + EMA3$	<code>3*EMA1 - 3*EMA2 + EMA3</code>	<code>prices</code> , <code>period</code>
166	<code>hma</code>	Hull Moving Average	WMA-based low-lag MA	<code>wma(2*wma(n/2) - wma(n), sqrt(n))</code>	<code>prices</code> , <code>period</code>
167	<code>kama</code>	Kaufman Adaptive MA	Efficiency-ratio adaptive MA	<code>KAMA_prev + SC*(Price - KAMA_prev)</code>	<code>prices</code> , <code>period</code>
168	<code>vwma</code>	Volume-Weighted MA	Price weighted by volume	<code>Sum(Price*Volume) / Sum(Volume)</code>	<code>prices</code> , <code>volumes</code> , <code>period</code>
169	<code>vwap</code>	Volume-Weighted Avg Price	Cumulative PV / cumulative V	<code>Sum(Typical_Price*Volume) / Sum(Volume)</code>	<code>highs</code> , <code>lows</code> , <code>closes</code> , <code>volumes</code>
170	<code>atr</code>	Average True Range	MA of true range	<code>MA(True_Range, n)</code>	<code>highs</code> , <code>lows</code> , <code>closes</code> , <code>period</code>
171	<code>true_range</code>	True Range	Max of HL, HC, LC	<code>max(H-L, abs(H-Cp), abs(L-Cp))</code>	<code>high</code> , <code>low</code> , <code>prev_close</code>
172	<code>bollinger_upper</code>	Bollinger Upper Band	$SMA + k \times std$	<code>SMA + 2*StdDev</code>	<code>prices</code> , <code>period</code>
173	<code>bollinger_lower</code>	Bollinger Lower Band	$SMA - k \times std$	<code>SMA - 2*StdDev</code>	<code>prices</code> , <code>period</code>
174	<code>bollinger_width</code>	Bollinger Band Width	$(Upper - Lower) / Middle$	<code>(Upper - Lower) / SMA</code>	<code>prices</code> , <code>period</code>
175	<code>bollinger_percent_b</code>	Bollinger %B	Position within bands	<code>(Price - Lower) / (Upper - Lower)</code>	<code>price</code> , <code>upper</code> , <code>lower</code>
176	<code>keltner_upper</code>	Keltner Upper Channel	$EMA + mult \times ATR$	<code>EMA + Mult*ATR</code>	<code>prices</code> , <code>highs</code> , <code>lows</code> , <code>period</code> , <code>multiplier</code>
177	<code>keltner_lower</code>	Keltner Lower Channel	$EMA - mult \times ATR$	<code>EMA - Mult*ATR</code>	<code>prices</code> , <code>highs</code> , <code>lows</code> , <code>period</code> , <code>multiplier</code>
178	<code>donchian_upper</code>	Donchian Upper Channel	Highest high over n	<code>max(Highs[-n:])</code>	<code>highs</code> , <code>period</code>
179	<code>donchian_lower</code>	Donchian Lower Channel	Lowest low over n	<code>min(Lows[-n:])</code>	<code>lows</code> , <code>period</code>
180	<code>donchian_middle</code>	Donchian Middle	Avg of upper and lower	<code>(Upper + Lower) / 2</code>	<code>highs</code> , <code>lows</code> , <code>period</code>
181	<code>rsi</code>	Relative Strength Index	Momentum oscillator 0-100	<code>100 - 100/(1 + AvgGain/AvgLoss)</code>	<code>prices</code> , <code>period</code>
182	<code>stochastic_k</code>	Stochastic %K	Position in high-low range	<code>(Close - LowN) / (HighN - LowN) * 100</code>	<code>highs</code> , <code>lows</code> , <code>closes</code> , <code>period</code>
183	<code>stochastic_d</code>	Stochastic %D	SMA of %K	<code>SMA(%K, 3)</code>	<code>stochastic_k</code> , <code>smoothing</code>
184	<code>macd_line</code>	MACD Line	$EMA12 - EMA26$	<code>EMA(12) - EMA(26)</code>	<code>prices</code>
185	<code>macd_signal</code>	MACD Signal Line	$EMA9$ of MACD	<code>EMA(MACD, 9)</code>	<code>macd_line</code>
186	<code>macd_histogram</code>	MACD Histogram	MACD - signal	<code>MACD_Line - Signal_Line</code>	<code>macd_line</code> , <code>signal_line</code>

#	ID	Name	Description	Formula	Inputs
187	cci	Commodity Channel Index	Deviation from typical price	$(TP - SMA_{TP}) / (0.015 * MeanDev)$	highs, lows, closes, period
188	williams_r	Williams %R	Inverse stochastic	$(HighN - Close) / (HighN - LowN) * -100$	highs, lows, closes, period
189	roc	Rate of Change %	Price momentum %	$(Price - Price_n) / Price_n * 100$	prices, period
190	momentum	Momentum	Price difference over n	$Price - Price_n$	prices, period
191	mfi	Money Flow Index	Volume-weighted RSI	$100 - 100 / (1 + PosFlow / NegFlow)$	highs, lows, closes, volumes, period
192	adx	Average Directional Index	Trend strength 0-100	$MA(DX, n)$	highs, lows, closes, period
193	plus_di	Plus Directional Indicator	Upward movement strength	$100 * EMA(+DM) / ATR$	highs, lows, closes, period
194	minus_di	Minus Directional Indicator	Downward movement strength	$100 * EMA(-DM) / ATR$	highs, lows, closes, period
195	aroon_up	Aroon Up	Periods since high	$(n - PeriodsSinceHigh) / n * 100$	highs, period
196	aroon_down	Aroon Down	Periods since low	$(n - PeriodsSinceLow) / n * 100$	lows, period
197	aroon_oscillator	Aroon Oscillator	Aroon up - down	$Aroon_{Up} - Aroon_{Down}$	highs, lows, period
198	parabolic_sar	Parabolic SAR	Stop and reverse trend	$SAR_{prev} + AF * (EP - SAR_{prev})$	highs, lows, acceleration
199	obv	On-Balance Volume	Cumulative volume flow	$Sum(Volume * Sign(Price_Change))$	closes, volumes
200	chaikin_money_flow	Chaikin Money Flow	Volume-weighted accumulation	$Sum(MFV) / Sum(Volume)$	highs, lows, closes, volumes, period
201	accumulation_distribution	Accumulation/ Distribution	Money flow volume cumulative	$Prev_AD + MFV$	highs, lows, closes, volumes
202	ichimoku_tenkan	Ichimoku Tenkan-sen	9-period midpoint	$(High9 + Low9) / 2$	highs, lows
203	ichimoku_kijun	Ichimoku Kijun-sen	26-period midpoint	$(High26 + Low26) / 2$	highs, lows
204	ichimoku_senkou_a	Ichimoku Senkou Span A	Avg of Tenkan and Kijun	$(Tenkan + Kijun) / 2$	tenkan, kijun
205	ichimoku_senkou_b	Ichimoku Senkou Span B	52-period midpoint	$(High52 + Low52) / 2$	highs, lows
206	linear_regression_slope	Linear Regression Slope	Trend slope of prices	Slope of best-fit line	prices, period
207	standard_deviation	Rolling Standard Deviation	Volatility measure	$sqrt(Sum((x - mean)^2) / n)$	prices, period
208	historical_volatility	Historical Volatility %	Annualized std of returns	$StdDev(LogReturns) * sqrt(252)$	prices, period
209	variance	Rolling Variance	Squared deviation	$Sum((x - mean)^2) / n$	prices, period
210	beta_coefficient	Beta Coefficient	Stock vs market sensitivity	$Cov(Stock, Market) / Var(Market)$	stock_returns, market_returns
211	correlation_coefficient	Correlation Coefficient	Linear relationship -1 to 1	$Cov(X, Y) / (StdX * StdY)$	series_x, series_y
212	z_score_price	Price Z-Score	Standardized price distance	$(Price - Mean) / StdDev$	prices, period
213	price_oscillator	Price Oscillator %	(Fast MA - Slow MA) / Slow	$(FastMA - SlowMA) / SlowMA * 100$	prices, fast, slow
214	trix	TRIX	Triple-smoothed ROC	ROC of Triple_EMA	prices, period
215	ultimate_oscillator	Ultimate Oscillator	Multi-timeframe momentum	$100 * Weighted_BP_Sum / TR_Sum$	highs, lows, closes
216	awesome_oscillator	Awesome Oscillator	SMA5 - SMA34 of midpoint	$SMA(MP, 5) - SMA(MP, 34)$	highs, lows
217	dpo	Detrended Price Oscillator	Price minus shifted SMA	$Price - SMA_{shifted}$	prices, period
218	vortex_positive	Vortex Indicator +VI	Upward trend movement	$Sum(+VM) / Sum(TR)$	highs, lows, closes, period
219	vortex_negative	Vortex Indicator -VI	Downward trend movement	$Sum(-VM) / Sum(TR)$	highs, lows, closes, period
220	mass_index	Mass Index	Range expansion reversal	$Sum(EMA9_HL / EMA9_EMA9_HL)$	highs, lows, period
221	force_index	Force Index	Price change x volume	$(Close - Prev_Close) * Volume$	closes, volumes
222	ease_of_movement	Ease of Movement	Price move per volume	$Distance_Moved / Box_Ratio$	highs, lows, volumes

#	ID	Name	Description	Formula	Inputs
223	klinger_oscillator	Klinger Oscillator	Volume force trend	$\text{EMA34}(\text{VF}) - \text{EMA55}(\text{VF})$	highs, lows, closes, volumes
224	chande_momentum	Chande Momentum Oscillator	Pure momentum -100 to 100	$(\text{Su} - \text{Sd}) / (\text{Su} + \text{Sd}) * 100$	prices, period
225	elder_ray_bull	Elder Ray Bull Power	High minus EMA	$\text{High} - \text{EMA}$	highs, prices, period
226	elder_ray_bear	Elder Ray Bear Power	Low minus EMA	$\text{Low} - \text{EMA}$	lows, prices, period
227	choppiness_index	Choppiness Index	Trend vs range 0-100	$100 * \log_{10}(\text{SumATR}/\text{Range}) / \log_{10}(n)$	highs, lows, closes, period
228	fisher_transform	Fisher Transform	Gaussian price normalizer	$0.5 * \ln((1+x)/(1-x))$	prices, period
229	coppock_curve	Coppock Curve	Long-term momentum	$\text{WMA10}(\text{ROC14} + \text{ROC11})$	prices
230	kst_oscillator	Know Sure Thing	Smoothed multi-ROC	$\text{Sum}(\text{weighted smoothed ROCs})$	prices
231	ppo	Percentage Price Oscillator	MACD as percentage	$(\text{EMA12} - \text{EMA26}) / \text{EMA26} * 100$	prices
232	pvo	Percentage Volume Oscillator	PPO applied to volume	$(\text{EMA12}_V - \text{EMA26}_V) / \text{EMA26}_V * 100$	volumes
233	relative_vigor_index	Relative Vigor Index	Close-open vs range	$\text{SMA}(\text{Close-Open}) / \text{SMA}(\text{High-Low})$	opens, highs, lows, closes
234	stochastic_rsi	Stochastic RSI	Stochastic of RSI	$(\text{RSI} - \text{MinRSI}) / (\text{MaxRSI} - \text{MinRSI})$	prices, period
235	supertrend	SuperTrend	ATR-based trend line	Based on ATR bands	highs, lows, closes, period, multiplier
236	pivot_point	Pivot Point	Floor trader pivot	$(\text{High} + \text{Low} + \text{Close}) / 3$	high, low, close
237	pivot_resistance_1	Pivot R1	First resistance	$2 * \text{Pivot} - \text{Low}$	pivot, low
238	pivot_support_1	Pivot S1	First support	$2 * \text{Pivot} - \text{High}$	pivot, high
239	fibonacci_retracement	Fibonacci Retracement	Key retracement levels	$\text{High} - (\text{High-Low}) * \text{Ratio}$	high, low, ratio
240	chandelier_exit_long	Chandelier Exit Long	ATR trailing stop	$\text{HighN} - \text{ATR} * \text{Multiplier}$	highs, lows, closes, period, multiplier

5.5 Options Pricing & Derivatives · [options.py](#) · 45 formulas

#	ID	Name	Description	Formula	Inputs
241	black_scholes_call	Black-Scholes Call	European call price	$S * N(d1) - K * e^{(-rT)} * N(d2)$	spot, strike, time, rate, volatility
242	black_scholes_put	Black-Scholes Put	European put price	$K * e^{(-rT)} * N(-d2) - S * N(-d1)$	spot, strike, time, rate, volatility
243	bs_d1	Black-Scholes d1	First BS parameter	$(\ln(S/K) + (r + \text{sig}^2/2)T) / (\text{sig} * \sqrt{T})$	spot, strike, time, rate, volatility
244	bs_d2	Black-Scholes d2	Second BS parameter	$d1 - \text{sig} * \sqrt{T}$	d1, volatility, time
245	bsm_call_dividend	BSM Call with Dividend	Call with continuous dividend	$S * e^{(-qT)} * N(d1) - K * e^{(-rT)} * N(d2)$	spot, strike, time, rate, volatility, dividend
246	bsm_put_dividend	BSM Put with Dividend	Put with continuous dividend	$K * e^{(-rT)} * N(-d2) - S * e^{(-qT)} * N(-d1)$	spot, strike, time, rate, volatility, dividend
247	delta_call	Delta (Call)	Price sensitivity to spot	$N(d1)$	spot, strike, time, rate, volatility
248	delta_put	Delta (Put)	Put price sensitivity	$N(d1) - 1$	spot, strike, time, rate, volatility
249	gamma	Gamma	Delta sensitivity to spot	$N'(d1) / (S * \text{sig} * \sqrt{T})$	spot, strike, time, rate, volatility
250	vega	Vega	Price sensitivity to volatility	$S * N'(d1) * \sqrt{T}$	spot, strike, time, rate, volatility
251	theta_call	Theta (Call)	Time decay of call	Time decay formula	spot, strike, time, rate, volatility
252	theta_put	Theta (Put)	Time decay of put	Time decay formula	spot, strike, time, rate, volatility
253	rho_call	Rho (Call)	Sensitivity to rate	$K * T * e^{(-rT)} * N(d2)$	spot, strike, time, rate, volatility
254	rho_put	Rho (Put)	Put rate sensitivity	$-K * T * e^{(-rT)} * N(-d2)$	spot, strike, time, rate, volatility
255	vanna	Vanna	Delta-vega cross sensitivity	$d(\text{Delta})/d(\text{vol})$	spot, strike, time, rate, volatility
256	charm	Charm	Delta decay	$d(\text{Delta})/d(\text{time})$	spot, strike, time, rate, volatility
257	vomma	Vomma	Vega convexity	$d(\text{Vega})/d(\text{vol})$	spot, strike, time, rate, volatility
258	speed	Speed	Gamma sensitivity to spot	$d(\text{Gamma})/d(S)$	spot, strike, time, rate, volatility
259	binomial_call	Binomial Call (CRR)	Tree-based call pricing	Cox-Ross-Rubinstein backward induction	spot, strike, time, rate, volatility, steps
260	binomial_put	Binomial Put (CRR)	Tree-based put pricing	CRR backward induction	spot, strike, time, rate, volatility, steps

#	ID	Name	Description	Formula	Inputs
261	trinomial_option	Trinomial Option	3-branch tree pricing	Trinomial backward induction	spot, strike, time, rate, volatility, steps
262	monte_carlo_option	Monte Carlo Option	Simulated path pricing	Mean(discounted payoffs)	spot, strike, time, rate, volatility, simulations
263	implied_volatility	Implied Volatility	Vol from market price	Newton-Raphson solve for sigma	option_price, spot, strike, time, rate
264	put_call_parity	Put-Call Parity	$C - P = S - Ke^{-(rT)}$	Call - Put = Spot - PV(Strike)	call, put, spot, strike, rate, time
265	intrinsic_value_call	Call Intrinsic Value	$\max(S - K, 0)$	max(Spot - Strike, 0)	spot, strike
266	intrinsic_value_put	Put Intrinsic Value	$\max(K - S, 0)$	max(Strike - Spot, 0)	spot, strike
267	time_value_option	Option Time Value	Price minus intrinsic	Option_Price - Intrinsic_Value	option_price, intrinsic_value
268	forward_price	Forward Price	Spot compounded to expiry	S*e^(rT)	spot, rate, time
269	futures_price	Futures Price	Spot with cost of carry	S*e^((r+storage-yield)T)	spot, rate, storage, yield, time
270	forward_rate_agreement	FRA Value	Forward rate agreement payoff	Notional*(Ref - FRA)*Days/360	notional, ref_rate, fra_rate, days
271	swap_fixed_rate	Swap Fixed Rate	Par swap rate	(1 - DF_n) / Sum(DF_i)	discount_factors
272	swap_value	Interest Rate Swap Value	PV fixed minus PV floating	PV_Fixed - PV_Floating	pv_fixed, pv_floating
273	call_payoff	Call Payoff at Expiry	Long call terminal value	max(S - K, 0) - Premium	spot, strike, premium
274	put_payoff	Put Payoff at Expiry	Long put terminal value	max(K - S, 0) - Premium	spot, strike, premium
275	straddle_payoff	Straddle Payoff	Long call + put	\ S - K\ - Total_Premium	spot, strike, total_premium
276	strangle_payoff	Strangle Payoff	OTM call + put	max(S-Kc,0)+max(Kp-S,0)-Prem	spot, strike_call, strike_put, premium
277	covered_call_return	Covered Call Return	Stock + short call yield	(Premium + max(K-S,0)) / S	spot, strike, premium
278	collar_value	Collar Value	Protective put + covered call	Long put + short call payoff	spot, put_strike, call_strike, net_premium
279	butterfly_payoff	Butterfly Spread Payoff	3-strike limited spread	Combined option payoff	spot, strikes, premiums
280	delta_hedge_shares	Delta Hedge Shares	Shares to hedge option	-Delta * Contracts * 100	delta, contracts
281	option_leverage	Option Leverage (Lambda)	Elasticity to underlying	Delta * S / Option_Price	delta, spot, option_price
282	breakeven_call	Call Breakeven	Strike plus premium	Strike + Premium	strike, premium
283	breakeven_put	Put Breakeven	Strike minus premium	Strike - Premium	strike, premium
284	max_pain	Max Pain Price	Strike with max option loss	Strike minimizing total payout	strikes, open_interest
285	historical_var_option	Option Position VaR	Value at risk for option book	Delta-gamma VaR approximation	delta, gamma, spot, volatility, confidence

5.6 Fixed Income & Bonds · [fixed_income.py](#) · 50 formulas

#	ID	Name	Description	Formula	Inputs
286	bond_price	Bond Price	PV of coupons + face	Sum(C/(1+y)^t) + F/(1+y)^n	coupon, face, yield, periods
287	bond_price_clean	Clean Bond Price	Dirty price minus accrued	Dirty_Price - Accrued_Interest	dirty_price, accrued_interest
288	bond_price_dirty	Dirty Bond Price	Clean plus accrued	Clean_Price + Accrued_Interest	clean_price, accrued_interest
289	accrued_interest	Accrued Interest	Coupon earned since last pay	Coupon * Days_Since / Days_Period	coupon, days_since, days_period
290	ytm	Yield to Maturity	IRR of bond cash flows	Solve y in price equation	price, coupon, face, periods
291	ytc	Yield to Call	Yield if called early	Solve y to call date	price, coupon, call_price, call_periods
292	ytw	Yield to Worst	Min of YTM and YTC	min(YTM, YTC)	ytm, ytc
293	current_yield	Current Yield	Annual coupon / price	Annual_Coupon / Price	annual_coupon, price
294	coupon_rate	Coupon Rate	Annual coupon / face	Annual_Coupon / Face_Value	annual_coupon, face_value
295	macaulay_duration	Macaulay Duration	Weighted avg time to CF	Sum(t*PV_CF) / Price	cash_flows, yield, periods
296	modified_duration	Modified Duration	Price sensitivity to yield	Macaulay / (1 + y/n)	macaulay_duration, yield, frequency
297	effective_duration	Effective Duration	Duration for embedded options	(P- - P+) / (2*P0*dy)	price_down, price_up, price_base, yield_change

#	ID	Name	Description	Formula	Inputs
298	dollar_duration	Dollar Duration	Price change per yield move	$\text{Modified_Duration} * \text{Price} * 0.0001$	modified_duration, price
299	convexity	Convexity	Curvature of price-yield	$\text{Sum}(t*(t+1)*\text{PV_CF}) / (\text{Price}*(1+y)^2)$	cash_flows, yield, periods
300	effective_convexity	Effective Convexity	Convexity with options	$(P_- + P_+ - 2*P_0) / (P_0*dy^2)$	price_down, price_up, price_base, yield_change
301	dv01	DV01 (PV01)	Dollar value of 1bp	$\text{Modified_Duration} * \text{Price} * 0.0001$	modified_duration, price
302	price_change_duration	Price Change (Duration)	Approx price move	$-\text{Modified_Duration} * \text{Price} * dy$	modified_duration, price, yield_change
303	price_change_convexity	Price Change (Dur+Conv)	Second-order price move	$-\text{MD}*P*dy + 0.5*\text{Conv}*P*dy^2$	modified_duration, convexity, price, yield_change
304	spot_rate	Spot Rate	Zero-coupon yield	$(\text{Face}/\text{Price})^{(1/n)} - 1$	price, face, periods
305	forward_rate	Forward Rate	Implied future rate	$((1+s_2)^{t_2}/(1+s_1)^{t_1})^{(1/(t_2-t_1))} - 1$	spot1, spot2, time1, time2
306	par_yield	Par Yield	Coupon for price = par	$(1 - \text{DF}_n) / \text{Sum}(\text{DF}_i)$	discount_factors
307	zero_coupon_price	Zero-Coupon Bond Price	PV of face value	$\text{Face} / (1+y)^n$	face, yield, periods
308	discount_factor	Discount Factor	PV of 1 unit	$1 / (1+r)^t$	rate, time
309	bond_equivalent_yield	Bond Equivalent Yield	Semi-annual to annual	$2 * ((1+y_{\text{semi}}) - 1)$	semi_annual_yield
310	effective_annual_yield	Effective Annual Yield	Compounded annual yield	$(1 + y/n)^n - 1$	yield, frequency
311	holding_period_return	Holding Period Return %	Total bond return	$(\text{End} + \text{Coupons} - \text{Start}) / \text{Start} * 100$	start_price, end_price, coupons
312	realized_compound_yield	Realized Compound Yield	Yield with reinvestment	$(\text{Total_FV} / \text{Price})^{(1/n)} - 1$	price, total_fv, periods
313	z_spread	Z-Spread	Constant spread over spot curve	Spread making PV = price	price, cash_flows, spot_rates
314	oas	Option-Adjusted Spread	Z-spread minus option cost	$\text{Z_Spread} - \text{Option_Cost}$	z_spread, option_cost
315	nominal_spread	Nominal Spread	Bond yield minus benchmark	$\text{Bond_YTM} - \text{Benchmark_YTM}$	bond_ytm, benchmark_ytm
316	g_spread	G-Spread	Spread over govt curve	$\text{Bond_Yield} - \text{Interpolated_Govt}$	bond_yield, govt_yield
317	i_spread	I-Spread	Spread over swap rate	$\text{Bond_Yield} - \text{Swap_Rate}$	bond_yield, swap_rate
318	asset_swap_spread	Asset Swap Spread	Spread in asset swap	Asset swap calculation	bond_price, coupon, swap_rate
319	credit_spread	Credit Spread	Risky minus risk-free yield	$\text{Corporate_Yield} - \text{Treasury_Yield}$	corporate_yield, treasury_yield
320	yield_curve_slope	Yield Curve Slope	Long minus short yield	$\text{Long_Yield} - \text{Short_Yield}$	long_yield, short_yield
321	yield_curve_butterfly	Yield Curve Butterfly	Curvature measure	$2*\text{Mid} - \text{Short} - \text{Long}$	short_yield, mid_yield, long_yield
322	key_rate_duration	Key Rate Duration	Sensitivity to one tenor	Price sensitivity to key rate	price_changes, yield_change
323	portfolio_duration	Portfolio Duration	Weighted avg duration	$\text{Sum}(\text{Weight}_i * \text{Duration}_i)$	weights, durations
324	portfolio_convexity	Portfolio Convexity	Weighted avg convexity	$\text{Sum}(\text{Weight}_i * \text{Convexity}_i)$	weights, convexities
325	reinvestment_income	Reinvestment Income	FV of reinvested coupons	$\text{Sum}(C*(1+r)^{(n-t)})$	coupon, rate, periods
326	interest_on_interest	Interest on Interest	Compounding of coupons	$\text{Reinvestment_Income} - \text{Total_Coupons}$	reinvestment_income, total_coupons
327	clean_to_invoice	Invoice Price	Clean x factor + accrued	$\text{Clean_Price}* \text{Factor} + \text{Accrued}$	clean_price, conversion_factor, accrued
328	bond_floor	Convertible Bond Floor	Straight bond value	PV of bond cash flows	coupon, face, yield, periods
329	conversion_value	Conversion Value	Shares x stock price	$\text{Conversion_Ratio} * \text{Stock_Price}$	conversion_ratio, stock_price
330	conversion_premium	Conversion Premium %	Bond price over conversion value	$(\text{Bond_Price} - \text{Conv_Value}) / \text{Conv_Value} * 100$	bond_price, conversion_value
331	tips_principal	TIPS Adjusted Principal	Inflation-adjusted face	$\text{Face} * \text{Index_Ratio}$	face, index_ratio
332	real_yield	Real Yield	Nominal minus inflation	$\text{Nominal_Yield} - \text{Inflation_Rate}$	nominal_yield, inflation_rate
333	breakeven_inflation	Breakeven Inflation Rate	Nominal minus real yield	$\text{Nominal_Yield} - \text{Real_Yield}$	nominal_yield, real_yield

#	ID	Name	Description	Formula	Inputs
334	rolling_yield	Rolling Yield (Carry+Roll)	Carry plus rolldown	Carry + Rolldown	carry, rolldown
335	expected_loss	Expected Loss (Credit)	PD x LGD x EAD	PD * LGD * EAD	pd, lgd, ead

5.7 Risk Management & Portfolio Theory · risk.py · 50 formulas

#	ID	Name	Description	Formula	Inputs
336	var_historical	VaR (Historical)	Loss at percentile	Percentile>Returns, 1-conf)	returns, confidence
337	var_parametric	VaR (Parametric)	Normal distribution VaR	-(mu + z*sigma)*Value	mean, std, confidence, value
338	var_monte_carlo	VaR (Monte Carlo)	Simulated loss distribution	Percentile of simulated P&L	returns, confidence, simulations
339	cvar	Conditional VaR (ES)	Expected loss beyond VaR	Mean(Losses > VaR)	returns, confidence
340	expected_shortfall	Expected Shortfall	Average tail loss	Mean>Returns below VaR)	returns, confidence
341	sharpe_ratio	Sharpe Ratio	Excess return per risk	(Return - Rf) / StdDev	returns, risk_free, std
342	sortino_ratio	Sortino Ratio	Return per downside risk	(Return - Rf) / Downside_Dev	returns, risk_free, downside_deviation
343	treynor_ratio	Treynor Ratio	Return per beta	(Return - Rf) / Beta	returns, risk_free, beta
344	information_ratio	Information Ratio	Active return per tracking error	(Return - Benchmark) / Tracking_Error	returns, benchmark, tracking_error
345	jensens_alpha	Jensen's Alpha	Return above CAPM	Return - (Rf + Beta*(Rm-Rf))	returns, risk_free, beta, market_return
346	calmar_ratio	Calmar Ratio	Return per max drawdown	Annual_Return / Max_Drawdown	annual_return, max_drawdown
347	sterling_ratio	Sterling Ratio	Return per avg drawdown	Annual_Return / Avg_Drawdown	annual_return, avg_drawdown
348	max_drawdown	Maximum Drawdown %	Largest peak-trough drop	(Trough - Peak) / Peak * 100	prices
349	drawdown_duration	Drawdown Duration	Time underwater	Periods from peak to recovery	prices
350	beta	Beta	Systematic risk vs market	Cov(Stock,Market) / Var(Market)	stock_returns, market_returns
351	alpha	Alpha	Excess return vs benchmark	Return - Benchmark_Return	returns, benchmark_return
352	tracking_error	Tracking Error	Std of active returns	StdDev(Portfolio - Benchmark)	portfolio_returns, benchmark_returns
353	downside_deviation	Downside Deviation	Std of negative returns	sqrt(Mean(min(0, R-MAR)^2))	returns, min_acceptable_return
354	semi_variance	Semi-Variance	Variance of downside	Mean((min(0, R-mean))^2)	returns
355	covariance	Covariance	Joint variability	Mean((X-Xbar)(Y-Ybar))	series_x, series_y
356	correlation	Correlation	Normalized covariance	Cov(X,Y) / (StdX*StdY)	series_x, series_y
357	portfolio_return	Portfolio Return	Weighted asset returns	Sum(Weight_i * Return_i)	weights, returns
358	portfolio_variance	Portfolio Variance	Markowitz 2-asset variance	w1^2*v1 + w2^2*v2 + 2*w1*w2*cov	weights, variances, covariance
359	portfolio_std	Portfolio Std Dev	Root of portfolio variance	sqrt(Portfolio_Variance)	portfolio_variance
360	portfolio_beta	Portfolio Beta	Weighted average beta	Sum(Weight_i * Beta_i)	weights, betas
361	minimum_variance_weight	Min Variance Weight	Optimal 2-asset weight	(v2-cov)/(v1+v2-2*cov)	variance1, variance2, covariance
362	efficient_frontier_return	Efficient Frontier Return	Optimal return for risk	Quadratic optimization	returns, covariance_matrix, target_risk
363	capital_allocation_line	Capital Allocation Line	Risk-return tradeoff line	Rf + Sharpe * Sigma	risk_free, sharpe, sigma
364	capital_market_line	Capital Market Line	Efficient frontier with Rf	Rf + ((Rm-Rf)/SigmaM)*Sigma	risk_free, market_return, market_std, portfolio_std

#	ID	Name	Description	Formula	Inputs
365	security_market_line	Security Market Line	CAPM expected return line	$R_f + \text{Beta} * (R_m - R_f)$	risk_free, beta, market_return
366	diversification_ratio	Diversification Ratio	Weighted vol / portfolio vol	$\text{Sum}(w * \text{sigma}) / \text{Portfolio_Sigma}$	weights, volatilities, portfolio_std
367	risk_parity_weight	Risk Parity Weight	Equal risk contribution	Inverse vol weighting	volatilities
368	marginal_var	Marginal VaR	VaR change per position	$d(\text{VaR})/d(\text{weight})$	weights, covariance_matrix, position
369	component_var	Component VaR	Position contribution to VaR	$\text{Marginal_VaR} * \text{Position}$	marginal_var, position_value
370	incremental_var	Incremental VaR	VaR change adding position	$\text{VaR_with} - \text{VaR_without}$	var_with, var_without
371	ulcer_index	Ulcer Index	Downside volatility measure	$\sqrt{\text{Mean}(\text{Drawdown}^2)}$	prices
372	gain_to_pain	Gain to Pain Ratio	Sum returns / sum losses	$\text{Sum}(\text{Returns}) / \text{abs}(\text{Sum}(\text{Losses}))$	returns
373	omega_ratio	Omega Ratio	Gains vs losses ratio	$\text{Sum}(\text{Gains}) / \text{Sum}(\text{Losses}) \text{ above threshold}$	returns, threshold
374	kappa_ratio	Kappa Ratio	Higher-moment downside ratio	$(\text{Return} - \text{MAR}) / \text{LPM}^{(1/n)}$	returns, min_acceptable_return, order
375	upside_potential_ratio	Upside Potential Ratio	Upside vs downside	$\text{Upside} / \text{Downside_Deviation}$	returns, min_acceptable_return
376	value_at_risk_normal	Parametric VaR (Normal)	Z-score based VaR	$\text{Value} * z * \text{sigma} * \sqrt{t}$	value, confidence, sigma, time
377	conditional_drawdown	Conditional Drawdown at Risk	Expected tail drawdown	Mean of worst drawdowns	prices, confidence
378	pain_index	Pain Index	Average drawdown depth	$\text{Mean}(\text{Drawdowns})$	prices
379	burke_ratio	Burke Ratio	Return per sqrt sum drawdowns	$\text{Return} / \sqrt{\text{Sum}(\text{DD}^2)}$	returns, drawdowns
380	m2_measure	M-Squared (M2)	Risk-adjusted vs market	$R_f + \text{Sharpe} * \text{Market_Std}$	sharpe, market_std, risk_free
381	active_premium	Active Premium	Return over benchmark	$\text{Annual_Return} - \text{Benchmark_Return}$	annual_return, benchmark_return
382	hurst_exponent	Hurst Exponent	Trend persistence measure	Rescaled range analysis	prices
383	kelly_criterion	Kelly Criterion	Optimal bet fraction	$(p * b - q) / b$	win_prob, win_loss_ratio
384	risk_of_ruin	Risk of Ruin	Probability of bankruptcy	$((1 - \text{edge}) / (1 + \text{edge}))^{\text{units}}$	edge, capital_units
385	expected_value	Expected Value	Probability-weighted outcome	$\text{Sum}(\text{Probability}_i * \text{Outcome}_i)$	probabilities, outcomes

5.8 Time Value of Money & Capital Budgeting · [tvm.py](#) · 45 formulas

#	ID	Name	Description	Formula	Inputs
386	present_value	Present Value	Discounted future value	$\text{FV} / (1+r)^n$	future_value, rate, periods
387	future_value	Future Value	Compounded present value	$\text{PV} * (1+r)^n$	present_value, rate, periods
388	npv	Net Present Value	Sum of discounted CFs minus cost	$\text{Sum}(\text{CF}_t / (1+r)^t) - \text{Initial}$	cash_flows, rate, initial_investment
389	irr	Internal Rate of Return	Rate where NPV = 0	Solve r: $\text{NPV} = 0$	cash_flows, initial_investment
390	mirr	Modified IRR	IRR with reinvestment rate	$(\text{FV_inflows} / \text{PV_outflows})^{(1/n)} - 1$	cash_flows, finance_rate, reinvest_rate
391	xirr	XIRR (Irregular)	IRR for irregular dates	Solve for rate with dates	cash_flows, dates
392	pv_annuity	PV of Annuity	PV of equal payments	$\text{PMT} * (1 - (1+r)^{-n}) / r$	payment, rate, periods
393	fv_annuity	FV of Annuity	FV of equal payments	$\text{PMT} * ((1+r)^n - 1) / r$	payment, rate, periods
394	pv_annuity_due	PV of Annuity Due	PV with payments at start	$\text{PV_Annuity} * (1+r)$	payment, rate, periods
395	fv_annuity_due	FV of Annuity Due	FV with payments at start	$\text{FV_Annuity} * (1+r)$	payment, rate, periods
396	perpetuity	Perpetuity Value	Infinite equal payments	PMT / r	payment, rate

#	ID	Name	Description	Formula	Inputs
397	growing_perpetuity	Growing Perpetuity	Infinite growing payments	$PMT / (r - g)$	payment, rate, growth
398	growing_annuity_pv	PV Growing Annuity	PV of growing payments	$PMT / (r - g) * (1 - ((1 + g) / (1 + r))^n)$	payment, rate, growth, periods
399	annuity_payment	Annuity Payment (PMT)	Payment for loan/annuity	$PV * r / (1 - (1 + r)^{-n})$	present_value, rate, periods
400	loan_payment	Loan Payment	Amortizing loan payment	$P * r / (1 - (1 + r)^{-n})$	principal, rate, periods
401	loan_balance	Remaining Loan Balance	Outstanding after k payments	$P * (1 + r)^k - PMT * ((1 + r)^k - 1) / r$	principal, rate, payment, periods_paid
402	amortization_interest	Amortization Interest Portion	Interest in payment k	$Balance * Rate$	balance, rate
403	amortization_principal	Amortization Principal Portion	Principal in payment k	$Payment - Interest$	payment, interest
404	effective_annual_rate	Effective Annual Rate	Annual compounded rate	$(1 + r/n)^n - 1$	nominal_rate, frequency
405	nominal_rate	Nominal Rate from EAR	APR from effective rate	$n * ((1 + EAR)^{1/n} - 1)$	ear, frequency
406	continuous_compounding	Continuous Compounding FV	FV with continuous rate	$PV * e^{(rt)}$	present_value, rate, time
407	continuous_pv	Continuous Compounding PV	PV with continuous rate	$FV * e^{(-rt)}$	future_value, rate, time
408	rule_of_72	Rule of 72	Doubling time estimate	$72 / Rate_Percent$	rate_percent
409	rule_of_69	Rule of 69.3	Continuous doubling time	$69.3 / Rate_Percent$	rate_percent
410	payback_period	Payback Period	Time to recover investment	Years until cumulative CF = 0	cash_flows, initial_investment
411	discounted_payback	Discounted Payback Period	Payback with discounting	Years until discounted cum = 0	cash_flows, rate, initial_investment
412	profitability_index	Profitability Index	PV inflows / initial cost	$PV_Inflows / Initial_Investment$	pv_inflows, initial_investment
413	equivalent_annual_cost	Equivalent Annual Cost	Annualized project cost	$NPV / Annuity_Factor$	npv, rate, periods
414	equivalent_annual_annuity	Equivalent Annual Annuity	Annualized NPV	$NPV * r / (1 - (1 + r)^{-n})$	npv, rate, periods
415	crossover_rate	Crossover Rate	Rate where two NPVs equal	IRR of CF differences	cash_flows_a, cash_flows_b
416	real_rate	Real Rate of Return	Inflation-adjusted rate	$(1 + nominal) / (1 + inflation) - 1$	nominal_rate, inflation_rate
417	fisher_equation	Fisher Equation	Nominal = real + inflation	$(1 + real) * (1 + inflation) - 1$	real_rate, inflation_rate
418	annuity_factor	Annuity Factor	PV factor for annuity	$(1 - (1 + r)^{-n}) / r$	rate, periods
419	future_value_factor	Future Value Factor	Compounding factor	$(1 + r)^n$	rate, periods
420	present_value_factor	Present Value Factor	Discount factor	$1 / (1 + r)^n$	rate, periods
421	sinking_fund	Sinking Fund Payment	Payment to reach FV	$FV * r / ((1 + r)^n - 1)$	future_value, rate, periods
422	capital_recovery_factor	Capital Recovery Factor	Annuity from present value	$r / (1 - (1 + r)^{-n})$	rate, periods
423	deferred_annuity_pv	Deferred Annuity PV	PV of delayed annuity	$PV_Annuity / (1 + r)^{defer}$	payment, rate, periods, deferral
424	net_future_value	Net Future Value	FV of all cash flows	$Sum(CF_t * (1 + r)^{(n - t)})$	cash_flows, rate
425	modified_payback	Modified Payback	Payback with terminal value	Adjusted payback period	cash_flows, terminal_value
426	accounting_rate_of_return	Accounting Rate of Return	Avg profit / avg investment	$Avg_Profit / Avg_Investment * 100$	average_profit, average_investment
427	bcr	Benefit-Cost Ratio	PV benefits / PV costs	$PV_Benefits / PV_Costs$	pv_benefits, pv_costs
428	annualized_return	Annualized Return	Geometric annual return	$(End / Start)^{(1 / years)} - 1$	start_value, end_value, years
429	holding_period_yield	Holding Period Yield %	Total period return	$(End + Income - Start) / Start * 100$	start_value, end_value, income
430	breakeven_interest_rate	Breakeven Interest Rate	Rate for indifference	Rate where NPV = 0	cash_flows

5.9 Corporate Finance & M&A · corporate_ma.py · 50 formulas

#	ID	Name	Description	Formula	Inputs
431	sustainable_growth_rate	Sustainable Growth Rate	ROE x retention	$ROE * Retention_Ratio$	roe, retention_ratio
432	internal_growth_rate	Internal Growth Rate	Growth without external finance	$(ROA * b) / (1 - ROA * b)$	roa, retention_ratio

#	ID	Name	Description	Formula	Inputs
433	plowback_ratio	Plowback Ratio	Retained earnings fraction	$1 - \text{Dividend_Payout}$	dividend_payout
434	roic	Return on Invested Capital	NOPAT / invested capital	$\text{NOPAT} / \text{Invested_Capital}$	nopat, invested_capital
435	invested_capital	Invested Capital	Debt + equity - cash	$\text{Total_Debt} + \text{Equity} - \text{Cash}$	total_debt, equity, cash
436	economic_profit	Economic Profit	NOPAT minus capital charge	$\text{NOPAT} - \text{Invested_Capital} * \text{WACC}$	nopat, invested_capital, wacc
437	hamada_equation	Hamada Equation	Levered beta from unlevered	$\text{Bu} * (1 + (1-T) * D/E)$	unlevered_beta, tax_rate, debt, equity
438	unlever_beta	Unlevered Beta	Asset beta from equity beta	$\text{Be} / (1 + (1-T) * D/E)$	levered_beta, tax_rate, debt, equity
439	relever_beta	Relevered Beta	Equity beta from asset beta	$\text{Ba} * (1 + (1-T) * D/E)$	asset_beta, tax_rate, debt, equity
440	mm_proposition1_no_tax	M&M Proposition I (No Tax)	Value independent of capital	$\text{VL} = \text{VU}$	unlevered_value
441	mm_proposition1_tax	M&M Proposition I (Tax)	Value with tax shield	$\text{VU} + \text{Tax_Rate} * \text{Debt}$	unlevered_value, tax_rate, debt
442	mm_proposition2	M&M Proposition II	Cost of equity with leverage	$\text{Ru} + (\text{Ru} - \text{Rd}) * (D/E) * (1-T)$	unlevered_cost, cost_debt, debt, equity, tax_rate
443	tax_shield	Tax Shield Value	PV of interest tax savings	$\text{Tax_Rate} * \text{Debt}$	tax_rate, debt
444	interest_tax_shield_annual	Annual Interest Tax Shield	Yearly tax savings	$\text{Interest} * \text{Tax_Rate}$	interest, tax_rate
445	degree_total_leverage	Degree of Total Leverage	Combined operating + financial	$\text{DOL} * \text{DFL}$	dol, dfl
446	free_cash_flow_firm	FCFF (Detailed)	Cash to all capital providers	$\text{NI} + \text{NCC} + \text{Int} * (1-T) - \text{FCInv} - \text{WCInv}$	net_income, ncc, interest, tax_rate, fcinv, wcinv
447	free_cash_flow_equity	FCFE (Detailed)	Cash to equity holders	$\text{NI} + \text{NCC} - \text{FCInv} - \text{WCInv} + \text{Net_Borrowing}$	net_income, ncc, fcinv, wcinv, net_borrowing
448	cash_flow_available_debt	CFADS	Cash available for debt service	$\text{EBITDA} - \text{Tax} - \text{WC} - \text{CapEx}$	ebitda, tax, working_capital_change, capex
449	accretion_dilution	Accretion/Dilution %	EPS change from deal	$(\text{ProForma_EPS} - \text{Standalone_EPS}) / \text{Standalone_EPS} * 100$	proforma_eps, standalone_eps
450	exchange_ratio	Exchange Ratio	Target shares per acquirer share	$\text{Offer_Price} / \text{Acquirer_Price}$	offer_price, acquirer_price
451	acquisition_premium	Acquisition Premium %	Offer over market price	$(\text{Offer} - \text{Market}) / \text{Market} * 100$	offer_price, market_price
452	synergy_value	Synergy Value	Combined value minus standalone	$\text{V_Combined} - (\text{V_A} + \text{V_B})$	combined_value, value_a, value_b
453	goodwill	Goodwill	Purchase price over fair value	$\text{Purchase_Price} - \text{Fair_Value_Net_Assets}$	purchase_price, fair_value_net_assets
454	purchase_price_allocation	Net Identifiable Assets	Fair value of assets minus liabilities	$\text{FV_Assets} - \text{FV_Liabilities}$	fv_assets, fv_liabilities
455	pro_forma_eps	Pro Forma EPS	Combined EPS post-merger	$\text{Combined_NI} / \text{Combined_Shares}$	combined_net_income, combined_shares
456	breakeven_synergies	Breakeven Synergies	Synergies to justify premium	$\text{Premium_Paid value}$	premium, target_shares
457	lbo_equity_return	LBO Equity Return (MOIC)	Multiple of invested capital	$\text{Exit_Equity} / \text{Entry_Equity}$	exit_equity, entry_equity
458	lbo_irr	LBO IRR	Annualized LBO return	$(\text{Exit/Entry})^{(1/\text{years})} - 1$	entry_equity, exit_equity, years
459	debt_paydown	Debt Paydown	Cumulative debt reduction	$\text{Entry_Debt} - \text{Exit_Debt}$	entry_debt, exit_debt
460	entry_multiple	Entry Multiple	Purchase EV / EBITDA	$\text{Entry_EV} / \text{EBITDA}$	entry_ev, ebitda
461	exit_multiple	Exit Multiple	Sale EV / EBITDA	$\text{Exit_EV} / \text{EBITDA}$	exit_ev, ebitda
462	sources_uses_balance	Sources and Uses Balance	Total sources = total uses	$\text{Sum}(\text{Sources}) - \text{Sum}(\text{Uses})$	sources, uses
463	net_borrowing	Net Borrowing	New debt minus repayments	$\text{Debt_Issued} - \text{Debt_Repaid}$	debt_issued, debt_repaid
464	dividend_discount_value	DDM Value	PV of all dividends	$\text{Sum}(D_t / (1+r)^t)$	dividends, rate

#	ID	Name	Description	Formula	Inputs
465	clientele_effect	After-Tax Dividend	Dividend net of tax	$\text{Dividend} * (1 - \text{Tax_Rate})$	dividend, tax_rate
466	share_buyback_eps_impact	Buyback EPS Impact	EPS after share reduction	$\text{NI} / (\text{Shares} - \text{Bought})$	net_income, shares, shares_bought
467	treasury_stock_method	Treasury Stock Method	Diluted shares from options	$\text{Options} - (\text{Options} * \text{Strike} / \text{Price})$	options, strike, price
468	weighted_avg_shares	Weighted Average Shares	Time-weighted share count	$\text{Sum}(\text{Shares}_i * \text{Months}_i / 12)$	share_periods
469	capital_structure_weight_equity	Equity Weight	Equity / total capital	$\text{Equity} / (\text{Equity} + \text{Debt})$	equity, debt
470	capital_structure_weight_debt	Debt Weight	Debt / total capital	$\text{Debt} / (\text{Equity} + \text{Debt})$	equity, debt
471	operating_working_capital	Operating Working Capital	Operating current assets - liabilities	$\text{Op_Current_Assets} - \text{Op_Current_Liabilities}$	op_current_assets, op_current_liabilities
472	invested_capital_turnover	Invested Capital Turnover	Revenue / invested capital	$\text{Revenue} / \text{Invested_Capital}$	revenue, invested_capital
473	reinvestment_rate	Reinvestment Rate	Net investment / NOPAT	$(\text{CapEx} - \text{Depr} + \text{WCInv}) / \text{NOPAT}$	capex, depreciation, wc_investment, nopat
474	expected_growth_fundamentals	Expected Growth (Fundamentals)	Reinvestment x ROIC	$\text{Reinvestment_Rate} * \text{ROIC}$	reinvestment_rate, roic
475	terminal_growth_implied	Implied Terminal Growth	Growth from terminal value	$r - \text{FCF} / \text{TV}$	rate, fcf, terminal_value
476	equity_value_from_ev	Equity Value from EV	EV minus net debt	$\text{EV} - \text{Net_Debt}$	ev, net_debt
477	net_debt_to_equity_value	Net Debt to Equity Value	Leverage in valuation	$\text{Net_Debt} / \text{Equity_Value}$	net_debt, equity_value
478	dilution_percentage	Dilution Percentage	New shares as % of total	$\text{New_Shares} / (\text{Old} + \text{New}) * 100$	old_shares, new_shares
479	control_premium	Control Premium %	Premium for control stake	$(\text{Control_Price} - \text{Minority}) / \text{Minority} * 100$	control_price, minority_price
480	minority_interest_value	Minority Interest Value	Non-controlling stake value	$\text{Subsidiary_Value} * \text{Minority_Pct}$	subsidiary_value, minority_pct

5.10 Accounting & Depreciation · [accounting.py](#) · 40 formulas

#	ID	Name	Description	Formula	Inputs
481	straight_line_depreciation	Straight-Line Depreciation	Equal annual depreciation	$(\text{Cost} - \text{Salvage}) / \text{Useful_Life}$	cost, salvage, useful_life
482	declining_balance	Declining Balance Depreciation	Accelerated depreciation	$\text{Book_Value} * \text{Rate}$	book_value, rate
483	double_declining_balance	Double Declining Balance	2x straight-line rate	$\text{Book_Value} * (2 / \text{Useful_Life})$	book_value, useful_life
484	units_of_production	Units of Production Depreciation	Depreciation per unit used	$(\text{Cost} - \text{Salvage}) / \text{Total_Units} * \text{Units_Used}$	cost, salvage, total_units, units_used
485	sum_of_years_digits	Sum-of-Years-Digits	Accelerated depreciation	$(\text{Cost} - \text{Salvage}) * \text{RemainingLife} / \text{SYD}$	cost, salvage, useful_life, current_year
486	macrs_depreciation	MACRS Depreciation	US tax depreciation	$\text{Cost} * \text{MACRS_Rate}$	cost, macrs_rate
487	accumulated_depreciation	Accumulated Depreciation	Total depreciation to date	$\text{Sum}(\text{Annual_Depreciation})$	annual_deprecations
488	book_value_asset	Net Book Value	Cost minus accumulated depr	$\text{Cost} - \text{Accumulated_Depreciation}$	cost, accumulated_depreciation
489	depreciation_rate	Depreciation Rate %	Annual depreciation / cost	$\text{Annual_Depreciation} / \text{Cost} * 100$	annual_depreciation, cost
490	amortization_intangible	Intangible Amortization	Straight-line for intangibles	$\text{Cost} / \text{Useful_Life}$	cost, useful_life
491	depletion	Depletion Expense	Natural resource expense	$(\text{Cost} - \text{Salvage}) / \text{Total_Units} * \text{Extracted}$	cost, salvage, total_units, units_extracted
492	fifo_cogs	FIFO COGS	First-in-first-out cost	$\text{Oldest inventory costs}$	inventory_layers, units_sold
493	lifo_cogs	LIFO COGS	Last-in-first-out cost	$\text{Newest inventory costs}$	inventory_layers, units_sold
494	weighted_average_cost	Weighted Average Cost	Average inventory cost	$\text{Total_Cost} / \text{Total_Units}$	total_cost, total_units

#	ID	Name	Description	Formula	Inputs
495	lifo_reserve	LIFO Reserve	FIFO minus LIFO inventory	$\text{FIFO_Inventory} - \text{LIFO_Inventory}$	fifo_inventory, lifo_inventory
496	inventory_write_down	Inventory Write-Down	Lower of cost or market	$\max(0, \text{Cost} - \text{Market})$	cost, market_value
497	ending_inventory	Ending Inventory	Beginning + purchases - COGS	$\text{Beginning} + \text{Purchases} - \text{COGS}$	beginning, purchases, cogs
498	cogs_calculation	COGS Calculation	Beginning + purchases - ending	$\text{Beginning} + \text{Purchases} - \text{Ending}$	beginning_inventory, purchases, ending_inventory
499	gross_profit_method	Gross Profit Method	Estimate inventory	$\text{Sales} - (\text{Sales} * \text{Gross_Margin})$	sales, gross_margin
500	bad_debt_percentage_sales	Bad Debt (% of Sales)	Bad debt from sales	$\text{Credit_Sales} * \text{Bad_Debt_Rate}$	credit_sales, bad_debt_rate
501	bad_debt_aging	Bad Debt (Aging)	Bad debt from receivables aging	$\text{Sum}(\text{Receivable_Bucket} * \text{Rate})$	receivable_buckets, rates
502	allowance_doubtful_accounts	Allowance for Doubtful Accounts	Estimated uncollectible	$\text{Receivables} * \text{Uncollectible_Rate}$	receivables, uncollectible_rate
503	net_realizable_value	Net Realizable Value	Receivables minus allowance	$\text{Receivables} - \text{Allowance}$	receivables, allowance
504	deferred_tax_liability	Deferred Tax Liability	Future tax on temp differences	$\text{Temp_Difference} * \text{Tax_Rate}$	temporary_difference, tax_rate
505	deferred_tax_asset	Deferred Tax Asset	Future tax benefit	$\text{Deductible_Difference} * \text{Tax_Rate}$	deductible_difference, tax_rate
506	effective_tax_rate_acct	Effective Tax Rate	Tax expense / pretax income	$\text{Tax_Expense} / \text{Pretax_Income} * 100$	tax_expense, pretax_income
507	stock_compensation_expense	Stock Comp Expense	Fair value amortized	$\text{Fair_Value} / \text{Vesting_Period}$	fair_value, vesting_period
508	pension_pbo	Projected Benefit Obligation	PV of pension obligations	$\text{PV of future benefits}$	benefits, discount_rate, periods
509	pension_funded_status	Pension Funded Status	Plan assets minus PBO	$\text{Plan_Assets} - \text{PBO}$	plan_assets, pbo
510	pension_expense	Net Periodic Pension Cost	Annual pension expense	$\text{Service} + \text{Interest} - \text{ExpReturn} + \text{Amort}$	service_cost, interest_cost, expected_return, amortization
511	operating_lease_expense	Operating Lease Expense	Straight-line lease cost	$\text{Total_Lease} / \text{Lease_Term}$	total_lease_payments, lease_term
512	finance_lease_liability	Finance Lease Liability	PV of lease payments	$\text{PV}(\text{Lease_Payments}, \text{Rate})$	lease_payments, rate, periods
513	right_of_use_asset	Right-of-Use Asset	Lease liability + costs	$\text{Lease_Liability} + \text{Initial_Costs}$	lease_liability, initial_costs
514	capitalized_interest	Capitalized Interest	Interest during construction	$\text{Avg_Expenditure} * \text{Rate}$	average_expenditure, interest_rate
515	revenue_recognition_percentage	Percentage of Completion	Revenue by completion	$\text{Total_Revenue} * \text{Pct_Complete}$	total_revenue, percent_complete
516	deferred_revenue	Deferred Revenue	Unearned revenue liability	$\text{Cash_Received} - \text{Revenue_Earned}$	cash_received, revenue_earned
517	comprehensive_income	Comprehensive Income	Net income + OCI	$\text{Net_Income} + \text{Other_Comprehensive_Income}$	net_income, oci
518	retained_earnings_ending	Ending Retained Earnings	Beginning + NI - dividends	$\text{Beginning_RE} + \text{Net_Income} - \text{Dividends}$	beginning_re, net_income, dividends
519	goodwill_impairment	Goodwill Impairment	Carrying minus fair value	$\max(0, \text{Carrying} - \text{Fair_Value})$	carrying_value, fair_value
520	asset_impairment	Asset Impairment Loss	Carrying over recoverable	$\max(0, \text{Carrying} - \text{Recoverable})$	carrying_value, recoverable_amount

5.11 Statistics & Econometrics · statistics_econ.py · 45 formulas

#	ID	Name	Description	Formula	Inputs
521	arithmetic_mean	Arithmetic Mean	Simple average	$\text{Sum}(x) / n$	values
522	geometric_mean	Geometric Mean	Compound average	$(\text{Prod}(x))^{(1/n)}$	values
523	harmonic_mean	Harmonic Mean	Reciprocal average	$n / \text{Sum}(1/x)$	values
524	weighted_mean	Weighted Mean	Weighted average	$\text{Sum}(w*x) / \text{Sum}(w)$	values, weights
525	median	Median	Middle value	$\text{Middle of sorted values}$	values

#	ID	Name	Description	Formula	Inputs
526	mode	Mode	Most frequent value	Most common value	values
527	range_stat	Range	Max minus min	Max - Min	values
528	variance_population	Population Variance	Average squared deviation	$\text{Sum}((x-\mu)^2) / N$	values
529	variance_sample	Sample Variance	Unbiased variance	$\text{Sum}((x-\bar{x})^2) / (n-1)$	values
530	standard_deviation_pop	Population Std Dev	Root of population variance	$\text{sqrt}(\text{Population_Variance})$	values
531	standard_deviation_sample	Sample Std Dev	Root of sample variance	$\text{sqrt}(\text{Sample_Variance})$	values
532	coefficient_variation	Coefficient of Variation	Std / mean	StdDev / Mean	values
533	skewness	Skewness	Distribution asymmetry	$E[(x-\mu)^3] / \sigma^3$	values
534	kurtosis	Kurtosis	Tail heaviness	$E[(x-\mu)^4] / \sigma^4$	values
535	excess_kurtosis	Excess Kurtosis	Kurtosis minus 3	Kurtosis - 3	values
536	covariance_stat	Covariance	Joint variability	$\text{Sum}((x-\bar{x})(y-\bar{y})) / (n-1)$	series_x, series_y
537	pearson_correlation	Pearson Correlation	Linear correlation	$\text{Cov}(X,Y) / (\text{SdX} * \text{SdY})$	series_x, series_y
538	spearman_correlation	Spearman Correlation	Rank correlation	$1 - 6 * \text{Sum}(d^2) / (n(n^2-1))$	series_x, series_y
539	linear_regression_beta	Regression Slope (Beta)	OLS slope coefficient	$\text{Cov}(X,Y) / \text{Var}(X)$	series_x, series_y
540	linear_regression_alpha	Regression Intercept	OLS intercept	$\bar{Y} - \text{Beta} * \bar{X}$	series_x, series_y
541	r_squared	R-Squared	Explained variance fraction	$1 - \text{SSres} / \text{SStot}$	actual, predicted
542	adjusted_r_squared	Adjusted R-Squared	R2 penalized for predictors	$1 - (1-R2)(n-1)/(n-k-1)$	r_squared, n, predictors
543	standard_error	Standard Error	Std of sampling distribution	$\text{StdDev} / \text{sqrt}(n)$	values
544	standard_error_regression	Standard Error of Regression	Residual standard error	$\text{sqrt}(\text{SSres} / (n-2))$	residuals, n
545	t_statistic	T-Statistic	Standardized test stat	$(\bar{x} - \mu) / (s / \text{sqrt}(n))$	sample_mean, pop_mean, std, n
546	z_score	Z-Score	Standard deviations from mean	$(x - \mu) / \sigma$	value, mean, std
547	confidence_interval	Confidence Interval	Range for parameter	$\text{mean} \pm z * (s / \text{sqrt}(n))$	mean, std, n, confidence
548	chi_square_stat	Chi-Square Statistic	Goodness of fit test	$\text{Sum}((O-E)^2 / E)$	observed, expected
549	f_statistic	F-Statistic	Variance ratio test	$\text{Var1} / \text{Var2}$	variance1, variance2
550	percentile	Percentile	Value at given percentile	Interpolated rank value	values, percentile
551	quartile	Quartile	25/50/75 split values	Percentile at 25/50/75	values, quartile_number
552	interquartile_range	Interquartile Range (IQR)	Q3 minus Q1	Q3 - Q1	values
553	autocorrelation	Autocorrelation	Self-correlation at lag	$\text{Corr}(x_t, x_{t-k})$	series, lag
554	moving_average_forecast	Moving Average Forecast	Average of last n	Mean(last n values)	values, window
555	exponential_smoothing	Exponential Smoothing	Weighted recent forecast	$\alpha * x + (1-\alpha) * \text{prev}$	values, alpha
556	holt_linear_trend	Holt's Linear Trend	Level + trend forecast	Level + Trend smoothing	values, alpha, beta
557	holt_winters	Holt-Winters Seasonal	Level + trend + seasonal	Triple exponential smoothing	values, alpha, beta, gamma, season_length
558	ar1_model	AR(1) Model	First-order autoregression	$c + \phi * x_{\text{prev}} + \text{error}$	series, phi, constant
559	durbin_watson	Durbin-Watson Statistic	Autocorrelation test	$\text{Sum}((e_t - e_{t-1})^2) / \text{Sum}(e_t^2)$	residuals
560	mean_absolute_error	Mean Absolute Error	Average absolute error	$\text{Mean}(\text{actual} - \text{predicted})$	actual, predicted
561	mean_squared_error	Mean Squared Error	Average squared error	$\text{Mean}((\text{actual} - \text{predicted})^2)$	actual, predicted
562	rmse	Root Mean Squared Error	Root of MSE	$\text{sqrt}(\text{MSE})$	actual, predicted
563	mape	Mean Absolute Percentage Error	Average % error	$\text{Mean}(\text{actual} - \text{pred} / \text{actual}) * 100$	actual, predicted
564	theil_u	Theil's U Statistic	Forecast accuracy	$\text{RMSE} / (\text{RMSE}_{\text{actual}} + \text{RMSE}_{\text{pred}})$	actual, predicted
565	garch_volatility	GARCH(1,1) Volatility	Conditional volatility	$\omega + \alpha * r^2 + \beta * \text{var}_{\text{prev}}$	returns, omega, alpha, beta

5.12 Core Math, Trigonometry, Linear Algebra & Geometry · `math_core.py` · 45 formulas

#	ID	Name	Description	Formula	Inputs
566	<code>sine</code>	Sine	Trig sine function	<code>sin(theta)</code>	angle_radians
567	<code>cosine</code>	Cosine	Trig cosine function	<code>cos(theta)</code>	angle_radians
568	<code>tangent</code>	Tangent	Trig tangent function	<code>tan(theta)</code>	angle_radians
569	<code>arcsine</code>	Arcsine	Inverse sine	<code>asin(x)</code>	value
570	<code>arccosine</code>	Arccosine	Inverse cosine	<code>acos(x)</code>	value
571	<code>arctangent</code>	Arctangent	Inverse tangent	<code>atan(x)</code>	value
572	<code>atan2</code>	Atan2	Two-argument arctangent	<code>atan2(y, x)</code>	y, x
573	<code>degrees_to_radians</code>	Degrees to Radians	Angle conversion	<code>deg * pi/180</code>	degrees
574	<code>radians_to_degrees</code>	Radians to Degrees	Angle conversion	<code>rad * 180/pi</code>	radians
575	<code>pythagorean</code>	Pythagorean Theorem	Right triangle hypotenuse	<code>sqrt(a^2 + b^2)</code>	a, b
576	<code>law_of_cosines</code>	Law of Cosines	Triangle side from angle	<code>sqrt(a^2+b^2-2ab*cos(C))</code>	a, b, angle_c
577	<code>law_of_sines</code>	Law of Sines	Triangle side-angle ratio	<code>a/sin(A) = b/sin(B)</code>	side_a, angle_a, angle_b
578	<code>hypotenuse</code>	Hypotenuse	Longest right-triangle side	<code>sqrt(a^2 + b^2)</code>	a, b
579	<code>euclidean_distance</code>	Euclidean Distance	Straight-line distance	<code>sqrt(Sum((x-y)^2))</code>	point_a, point_b
580	<code>manhattan_distance</code>	Manhattan Distance	Taxicab distance	<code>Sum(x-y)</code>	point_a, point_b
581	<code>cosine_similarity</code>	Cosine Similarity	Vector angle similarity	<code>A.B / (A * B)</code>	vector_a, vector_b
582	<code>minkowski_distance</code>	Minkowski Distance	Generalized distance	<code>(Sum(x-y ^p))^(1/p)</code>	point_a, point_b, p
583	<code>chebyshev_distance</code>	Chebyshev Distance	Max coordinate difference	<code>max(x-y)</code>	point_a, point_b
584	<code>mahalanobis_distance</code>	Mahalanobis Distance	Covariance-scaled distance	<code>sqrt((x-mu)' Cov^-1 (x-mu))</code>	point, mean, covariance_matrix
585	<code>hamming_distance</code>	Hamming Distance	Differing positions count	<code>Count(x != y)</code>	sequence_a, sequence_b
586	<code>dot_product</code>	Dot Product	Vector inner product	<code>Sum(a_i * b_i)</code>	vector_a, vector_b
587	<code>cross_product_2d</code>	Cross Product (2D)	Scalar cross product	<code>a_x*b_y - a_y*b_x</code>	vector_a, vector_b
588	<code>vector_magnitude</code>	Vector Magnitude	Euclidean norm	<code>sqrt(Sum(x^2))</code>	vector
589	<code>vector_normalize</code>	Vector Normalization	Unit vector	<code>v / v </code>	vector
590	<code>matrix_multiply</code>	Matrix Multiplication	Matrix product	<code>C_ij = Sum(A_ik * B_kj)</code>	matrix_a, matrix_b
591	<code>matrix_transpose</code>	Matrix Transpose	Flip rows/columns	<code>A_ij -> A_ji</code>	matrix
592	<code>matrix_determinant</code>	Matrix Determinant	Scalar from square matrix	<code>det(A)</code>	matrix
593	<code>matrix_inverse</code>	Matrix Inverse	Multiplicative inverse	<code>A^-1</code>	matrix
594	<code>matrix_trace</code>	Matrix Trace	Sum of diagonal	<code>Sum(A_ii)</code>	matrix
595	<code>eigenvalues</code>	Eigenvalues	Characteristic roots	<code>Solve det(A-lambda*I)=0</code>	matrix
596	<code>cholesky_decomposition</code>	Cholesky Decomposition	Lower triangular factor	<code>A = L*L'</code>	matrix
597	<code>logarithm_natural</code>	Natural Logarithm	Log base e	<code>ln(x)</code>	value
598	<code>logarithm_base10</code>	Log Base 10	Common logarithm	<code>log10(x)</code>	value
599	<code>logarithm_base</code>	Log Arbitrary Base	Log base b	<code>log(x) / log(b)</code>	value, base
600	<code>exponential</code>	Exponential	e to the power x	<code>e^x</code>	value
601	<code>power_function</code>	Power Function	Base to exponent	<code>base^exponent</code>	base, exponent
602	<code>nth_root</code>	Nth Root	Root of degree n	<code>x^(1/n)</code>	value, n
603	<code>factorial</code>	Factorial	Product 1 to n	<code>n!</code>	n
604	<code>combination</code>	Combination (nCr)	Ways to choose r from n	<code>n! / (r!(n-r)!)</code>	n, r
605	<code>permutation</code>	Permutation (nPr)	Ordered arrangements	<code>n! / (n-r)!</code>	n, r
606	<code>absolute_value</code>	Absolute Value	Magnitude without sign	<code> x </code>	value
607	<code>percentage_change</code>	Percentage Change	Relative change %	<code>(New - Old)/Old * 100</code>	old_value, new_value
608	<code>percentage_of_total</code>	Percentage of Total	Part over whole	<code>Part / Total * 100</code>	part, total
609	<code>compound_growth</code>	Compound Growth	Multi-period growth	<code>Initial*(1+r)^n</code>	initial, rate, periods
610	<code>cagr</code>	CAGR	Compound annual growth rate	<code>(End/Start)^(1/years) - 1</code>	start_value, end_value, years

5.13 Growth, Segment, Forensic & Modern Metrics · `growth_segment_forensic.py` · 50 formulas

#	ID	Name	Description	Formula	Inputs
611	<code>yoy_change_absolute</code>	YoY Change (Absolute)	Dollar change year over year	$\text{Current} - \text{Prior}$	current, prior
612	<code>yoy_change_pct</code>	YoY Change (%)	Percent change year over year	$(\text{Current} - \text{Prior}) / \text{Prior} * 100$	current, prior
613	<code>sequential_growth</code>	Sequential (QoQ) Growth %	Quarter over quarter change	$(\text{Current_Q} - \text{Prior_Q}) / \text{Prior_Q} * 100$	current_q, prior_q
614	<code>ttm</code>	Trailing Twelve Months	Sum of last four quarters	$Q1 + Q2 + Q3 + Q4$	q1, q2, q3, q4
615	<code>ttm_rolling</code>	TTM Rolling Update	New TTM = old TTM - oldest Q + newest Q	$\text{Prior_TTM} - \text{Dropped_Q} + \text{New_Q}$	prior_ttm, dropped_q, new_q
616	<code>quarter_annualized</code>	Quarterly Annualized Run-Rate	Quarter times four	$\text{Quarter_Value} * 4$	quarter_value
617	<code>monthly_annualized</code>	Monthly Annualized Run-Rate	Month times twelve	$\text{Monthly_Value} * 12$	monthly_value
618	<code>percentage_point_change</code>	Percentage Point Change	Difference of two percentages	$\text{Current_Pct} - \text{Prior_Pct}$	current_pct, prior_pct
619	<code>compound_quarterly</code>	Compound Sub-Annual Growth	Compounded periodic growth	$(1 + \text{periodic_rate})^{\text{periods}} - 1$	periodic_rate, periods
620	<code>constant_currency_growth</code>	Constant Currency Growth %	FX-neutral growth	$(\text{Current_CC} - \text{Prior}) / \text{Prior} * 100$	current_cc, prior
621	<code>organic_growth</code>	Organic Growth %	Growth excluding M&A and FX	$(\text{Reported_Growth} - \text{MA_Contribution} - \text{FX_Contribution})$	reported_growth, ma_contribution, fx_contribution
622	<code>inorganic_growth</code>	Inorganic Growth %	Growth from acquisitions	$\text{MA_Revenue} / \text{Prior_Revenue} * 100$	ma_revenue, prior_revenue
623	<code>two_year_stack</code>	Two-Year Stacked Growth %	Sum of consecutive YoY rates	$\text{Growth_Y1} + \text{Growth_Y2}$	growth_y1, growth_y2
624	<code>multi_year_cagr</code>	Multi-Year CAGR	Compound growth over n years	$(\text{End}/\text{Start})^{(1/\text{years})} - 1$	start, end, years
625	<code>multi_year_average</code>	Multi-Year Average	Mean across periods	$\text{Sum}(\text{values}) / \text{count}$	values
626	<code>dividend_growth_rate</code>	Dividend Growth Rate (CAGR)	Compound dividend growth	$(\text{D_end}/\text{D_start})^{(1/\text{years})} - 1$	d_start, d_end, years
627	<code>revenue_run_rate</code>	Revenue Run-Rate	Annualized current revenue	$\text{Current_Period_Revenue} * \text{Periods_Per_Year}$	current_period_revenue, periods_per_year
628	<code>segment_growth</code>	Segment Revenue Growth %	Per-segment YoY growth	$(\text{Seg_Current} - \text{Seg_Prior}) / \text{Seg_Prior} * 100$	seg_current, seg_prior
629	<code>segment_margin</code>	Segment Operating Margin %	Segment profit over segment revenue	$\text{Segment_Operating_Income} / \text{Segment_Revenue} * 100$	segment_operating_income, segment_revenue
630	<code>segment_revenue_share</code>	Segment Revenue Share %	Segment as % of total	$\text{Segment_Revenue} / \text{Total_Revenue} * 100$	segment_revenue, total_revenue
631	<code>segment_contribution</code>	Segment Profit Contribution %	Segment profit / total profit	$\text{Segment_Profit} / \text{Total_Profit} * 100$	segment_profit, total_profit
632	<code>mix_shift</code>	Revenue Mix Shift (pp)	Change in segment share	$\text{Current_Share_Pct} - \text{Prior_Share_Pct}$	current_share_pct, prior_share_pct
633	<code>geographic_concentration</code>	Geographic Concentration %	Region revenue / total	$\text{Region_Revenue} / \text{Total_Revenue} * 100$	region_revenue, total_revenue
634	<code>customer_concentration</code>	Customer Concentration %	Top customer revenue / total	$\text{Top_Customer_Revenue} / \text{Total_Revenue} * 100$	top_customer_revenue, total_revenue
635	<code>herfindahl_index</code>	Herfindahl Concentration Index	Sum of squared shares	$\text{Sum}(\text{Share_i}^2)$	shares
636	<code>weighted_segment_growth</code>	Weighted Segment Growth %	Share-weighted segment growth	$\text{Sum}(\text{Share_i} * \text{Growth_i})$	shares, growths
637	<code>beneish_m_score</code>	Beneish M-Score	Earnings manipulation detector	$-4.84 + \text{weighted } 8 \text{ indices}$	dsri, gmi, aqi, sgi, depi, sgai, lvgi, tata
638	<code>sloan_ratio</code>	Sloan Accrual Ratio %	Accruals / total assets	$(\text{NI} - \text{CF0} - \text{CFI}) / \text{Total_Assets} * 100$	net_income, cfo, cfi, total_assets
639	<code>accruals_ratio_bs</code>	Balance Sheet Accruals Ratio	Change in NOA / avg NOA	$(\text{NOA_end} - \text{NOA_start}) / \text{Avg_NOA}$	noa_end, noa_start

#	ID	Name	Description	Formula	Inputs
640	accruals_ratio_cf	Cash Flow Accruals Ratio	(NI - CFO - CFI)/avg NOA	$(NI - CFO - CFI) / Avg_NOA$	net_income, cfo, cfi, avg_noa
641	cash_conversion	Cash Conversion Ratio	Operating cash flow / net income	CFO / Net_Income	cfo, net_income
642	fcf_conversion	FCF Conversion %	FCF / net income	$FCF / Net_Income * 100$	fcf, net_income
643	earnings_quality_ratio	Earnings Quality Ratio	CFO / net income	CFO / Net_Income	cfo, net_income
644	adjusted_ebitda	Adjusted EBITDA	EBITDA plus one-time addbacks	$EBITDA + Addbacks$	ebitda, addbacks
645	normalized_earnings	Normalized Earnings	Net income excluding one-timers	$Net_Income - One_Time_Items$	net_income, one_time_items
646	days_cash_on_hand	Days Cash on Hand	Cash / daily operating expense	$Cash / (Operating_Expenses / 365)$	cash, operating_expenses
647	net_working_capital_change	Change in Net Working Capital	Period-over-period NWC change	$NWC_Current - NWC_Prior$	nwc_current, nwc_prior
648	capex_to_depreciation	CapEx to Depreciation	Reinvestment signal	$CapEx / Depreciation$	capex, depreciation
649	maintenance_capex_estimate	Maintenance CapEx (Est.)	Depreciation as maintenance proxy	$Depreciation$	depreciation
650	growth_capex	Growth CapEx	CapEx above maintenance	$CapEx - Maintenance_CapEx$	capex, maintenance_capex
651	incremental_roic	Incremental ROIC %	Change in NOPAT / change in capital	$Delta_NOPAT / Delta_Invested_Capital * 100$	delta_nopat, delta_invested_capital
652	cfroi	Cash Flow Return on Investment %	Gross cash flow / gross investment	$Gross_Cash_Flow / Gross_Investment * 100$	gross_cash_flow, gross_investment
653	buyback_yield	Buyback Yield %	Net buybacks / market cap	$Net_Buybacks / Market_Cap * 100$	net_buybacks, market_cap
654	total_payout_ratio	Total Payout Ratio %	Dividends plus buybacks / NI	$(Dividends + Buybacks) / Net_Income * 100$	dividends, buybacks, net_income
655	total_yield	Total Shareholder Yield %	Dividend yield plus buyback yield	$Dividend_Yield + Buyback_Yield$	dividend_yield, buyback_yield
656	effective_interest_rate	Effective Interest Rate %	Interest expense / avg debt	$Interest_Expense / Average_Debt * 100$	interest_expense, average_debt
657	weighted_avg_cost_debt	Weighted Avg Cost of Debt %	Sum of weighted debt rates	$Sum(Weight_i * Rate_i)$	weights, rates
658	arpu	Average Revenue Per User	Revenue / users	$Revenue / Users$	revenue, users
659	net_revenue_retention	Net Revenue Retention %	Expansion-adjusted retention	$(Start + Expansion - Churn - Contraction) / Start * 100$	starting_revenue, expansion, churn, contraction
660	ltv_cac_ratio	LTV/CAC Ratio	Lifetime value over acquisition cost	LTV / CAC	ltv, cac

5.14 AI / Machine Learning Metrics & Functions · [ai_ml.py](#) · 93 formulas

#	ID	Name	Description	Formula	Inputs
661	mse_loss	Mean Squared Error Loss	Average squared prediction error	$Mean((y - \hat{y})^2)$	y_true, y_pred
662	mae_loss	Mean Absolute Error Loss	Average absolute prediction error	$Mean(y - \hat{y})$	y_true, y_pred
663	rmse_loss	Root Mean Squared Error Loss	Root of MSE	$\sqrt{Mean((y - \hat{y})^2)}$	y_true, y_pred
664	huber_loss	Huber Loss	MSE/MAE hybrid robust to outliers	$0.5 * e^2 \text{ if } e \leq d \text{ else } d * (e - 0.5d)$	y_true, y_pred, delta
665	cross_entropy_loss	Cross-Entropy Loss	Multi-class log loss	$-Sum(y * \log(\hat{y}))$	y_true, y_pred
666	binary_cross_entropy	Binary Cross-Entropy	Two-class log loss	$-Mean(y * \log(p) + (1 - y) * \log(1 - p))$	y_true, y_pred
667	categorical_cross_entropy	Categorical Cross-Entropy	One-hot multi-class loss	$-Sum(y_i * \log(p_i))$	y_true, y_pred
668	hinge_loss	Hinge Loss	SVM margin loss	$Mean(\max(0, 1 - y * \hat{y}))$	y_true, y_pred
669	kl_divergence	KL Divergence	Distribution difference	$Sum(p * \log(p/q))$	p_dist, q_dist

#	ID	Name	Description	Formula	Inputs
670	<code>focal_loss</code>	Focal Loss	Class-imbalance weighted CE	$-\alpha \cdot (1-p)^{\gamma} \cdot \log(p)$	<code>y_true</code> , <code>y_pred</code> , <code>alpha</code> , <code>gamma</code>
671	<code>log_loss</code>	Log Loss	Logarithmic probability loss	$-\text{Mean}(y \cdot \log(p) + (1-y) \cdot \log(1-p))$	<code>y_true</code> , <code>y_pred</code>
672	<code>msle_loss</code>	Mean Squared Log Error	Log-scaled squared error	$\text{Mean}((\log(1+y) - \log(1+\hat{y}))^2)$	<code>y_true</code> , <code>y_pred</code>
673	<code>sigmoid</code>	Sigmoid	Logistic squashing 0-1	$1 / (1 + e^{-x})$	<code>x</code>
674	<code>relu</code>	ReLU	Rectified linear unit	$\max(0, x)$	<code>x</code>
675	<code>leaky_relu</code>	Leaky ReLU	ReLU with small negative slope	$x \text{ if } x > 0 \text{ else } \alpha \cdot x$	<code>x</code> , <code>alpha</code>
676	<code>tanh_activation</code>	Tanh	Hyperbolic tangent -1 to 1	$(e^x - e^{-x}) / (e^x + e^{-x})$	<code>x</code>
677	<code>softmax</code>	Softmax	Normalized exponential probabilities	$e^{x_i} / \sum(e^{x_j})$	<code>x_vector</code>
678	<code>gelu</code>	GELU	Gaussian error linear unit	$x \cdot \Phi(x)$	<code>x</code>
679	<code>elu</code>	ELU	Exponential linear unit	$x \text{ if } x > 0 \text{ else } \alpha \cdot (e^x - 1)$	<code>x</code> , <code>alpha</code>
680	<code>swish</code>	Swish	Self-gated activation	$x \cdot \text{sigmoid}(x)$	<code>x</code>
681	<code>softplus</code>	Softplus	Smooth ReLU	$\log(1 + e^x)$	<code>x</code>
682	<code>accuracy</code>	Accuracy	Correct / total predictions	$(TP + TN) / (TP + TN + FP + FN)$	<code>tp</code> , <code>tn</code> , <code>fp</code> , <code>fn</code>
683	<code>precision</code>	Precision	True positives / predicted positives	$TP / (TP + FP)$	<code>tp</code> , <code>fp</code>
684	<code>recall</code>	Recall (Sensitivity)	True positives / actual positives	$TP / (TP + FN)$	<code>tp</code> , <code>fn</code>
685	<code>f1_score</code>	F1 Score	Harmonic mean precision-recall	$2 \cdot P \cdot R / (P + R)$	<code>precision</code> , <code>recall</code>
686	<code>f_beta_score</code>	F-Beta Score	Weighted precision-recall mean	$(1 + \beta^2) \cdot P \cdot R / (\beta^2 \cdot P + R)$	<code>precision</code> , <code>recall</code> , <code>beta</code>
687	<code>specificity</code>	Specificity	True negatives / actual negatives	$TN / (TN + FP)$	<code>tn</code> , <code>fp</code>
688	<code>roc_auc</code>	ROC AUC	Area under ROC curve	Integral of TPR over FPR	<code>y_true</code> , <code>y_scores</code>
689	<code>pr_auc</code>	PR AUC	Area under precision-recall curve	Integral of precision over recall	<code>y_true</code> , <code>y_scores</code>
690	<code>matthews_corr</code>	Matthews Correlation Coefficient	Balanced binary quality	$(TP \cdot TN - FP \cdot FN) / \sqrt{...}$	<code>tp</code> , <code>tn</code> , <code>fp</code> , <code>fn</code>
691	<code>cohen_kappa</code>	Cohen's Kappa	Agreement beyond chance	$(p_o - p_e) / (1 - p_e)$	<code>observed_agreement</code> , <code>expected_agreement</code>
692	<code>balanced_accuracy</code>	Balanced Accuracy	Mean of recall per class	$(\text{Sensitivity} + \text{Specificity}) / 2$	<code>sensitivity</code> , <code>specificity</code>
693	<code>false_positive_rate</code>	False Positive Rate	FP / actual negatives	$FP / (FP + TN)$	<code>fp</code> , <code>tn</code>
694	<code>false_negative_rate</code>	False Negative Rate	FN / actual positives	$FN / (FN + TP)$	<code>fn</code> , <code>tp</code>
695	<code>r2_score</code>	R-Squared Score	Variance explained	$1 - SS_{res} / SS_{tot}$	<code>y_true</code> , <code>y_pred</code>
696	<code>adjusted_r2_ml</code>	Adjusted R-Squared	R2 penalized for features	$1 - (1 - R^2)(n-1)/(n-k-1)$	<code>r2</code> , <code>n</code> , <code>k</code>
697	<code>explained_variance</code>	Explained Variance Score	1 - Var(residual)/Var(y)	$1 - \text{Var}(y - \hat{y}) / \text{Var}(y)$	<code>y_true</code> , <code>y_pred</code>
698	<code>mape_metric</code>	MAPE	Mean absolute percent error	$\text{Mean}(y - \hat{y} / y) \cdot 100$	<code>y_true</code> , <code>y_pred</code>
699	<code>smape</code>	SMAPE	Symmetric MAPE	$\text{Mean}(2 \cdot y - \hat{y} / (y + \hat{y})) \cdot 100$	<code>y_true</code> , <code>y_pred</code>
700	<code>median_absolute_error</code>	Median Absolute Error	Median of absolute errors	$\text{Median}(y - \hat{y})$	<code>y_true</code> , <code>y_pred</code>
701	<code>jaccard_similarity</code>	Jaccard Similarity	Intersection over union	$ A \cap B / A \cup B $	<code>set_a</code> , <code>set_b</code>
702	<code>dice_coefficient</code>	Dice Coefficient	2x intersection over sum	$2 \cdot A \cap B / (A + B)$	<code>set_a</code> , <code>set_b</code>
703	<code>canberra_distance</code>	Canberra Distance	Weighted Manhattan	$\sum(x - y / (x + y))$	<code>vector_a</code> , <code>vector_b</code>
704	<code>braycurtis_distance</code>	Bray-Curtis Distance	Compositional dissimilarity	$\sum(x - y) / \sum(x + y)$	<code>vector_a</code> , <code>vector_b</code>
705	<code>haversine_distance</code>	Haversine Distance	Great-circle distance	$2r \cdot \arcsin(\sqrt{\text{hav}})$	<code>lat1</code> , <code>lon1</code> , <code>lat2</code> , <code>lon2</code>
706	<code>jaro_winkler</code>	Jaro-Winkler Similarity	String similarity with prefix	$\text{Jaro} + \text{prefix} \cdot \text{scale} \cdot (1 - \text{Jaro})$	<code>string_a</code> , <code>string_b</code>

#	ID	Name	Description	Formula	Inputs
707	silhouette_score	Silhouette Score	Cluster cohesion vs separation	$(b - a) / \max(a, b)$	intra_distance, nearest_cluster_distance
708	davies_bouldin	Davies-Bouldin Index	Avg cluster similarity	$\text{Mean}(\max((s_i + s_j) / d_{ij}))$	cluster_scatters, cluster_distances
709	calinski_harabasz	Calinski-Harabasz Index	Between/within variance ratio	$(BGSS/WGSS) * ((n - k) / (k - 1))$	between_ss, within_ss, n, k
710	inertia	Inertia (WCSS)	Within-cluster sum of squares	$\text{Sum}(\ x - \text{centroid}\ ^2)$	points, centroids
711	dunn_index	Dunn Index	Min inter / max intra cluster	$\text{Min_Inter_Cluster} / \text{Max_Intra_Cluster}$	inter_distances, intra_distances
712	rand_index	Rand Index	Clustering agreement	$(a + b) / C(n, 2)$	agreements, n
713	adjusted_rand_index	Adjusted Rand Index	Chance-corrected Rand	$(RI - \text{Expected}) / (\text{Max} - \text{Expected})$	contingency_table
714	normalized_mutual_info	Normalized Mutual Information	Clustering MI normalized	$MI / \sqrt{H(U) * H(V)}$	labels_true, labels_pred
715	entropy	Shannon Entropy	Information content	$-\text{Sum}(p * \log_2(p))$	probabilities
716	conditional_entropy	Conditional Entropy	Entropy given another variable	$H(Y) - I(X; Y)$	joint_dist, marginal_dist
717	mutual_information	Mutual Information	Shared information	$\text{Sum}(p * \log(p / (p_x * p_y)))$	joint_dist, marginal_x, marginal_y
718	information_gain	Information Gain	Entropy reduction from split	$H(\text{parent}) - \text{Weighted_H}(\text{children})$	parent_entropy, child_entropies, weights
719	gini_impurity	Gini Impurity	Node impurity for trees	$1 - \text{Sum}(p^2)$	class_probabilities
720	gain_ratio	Gain Ratio	Information gain / split info	$\text{Information_Gain} / \text{Split_Info}$	information_gain, split_info
721	gradient_descent_step	Gradient Descent Step	Parameter update rule	$\theta - \text{lr} * \text{gradient}$	theta, learning_rate, gradient
722	momentum_update	Momentum Update	Velocity-based update	$\beta * v + (1 - \beta) * \text{gradient}$	velocity, gradient, beta
723	adam_update	Adam Optimizer Step	Adaptive moment estimation	$\theta - \text{lr} * \hat{m} / (\sqrt{\hat{v}} + \epsilon)$	theta, m_hat, v_hat, learning_rate, epsilon
724	rmsprop_update	RMSProp Update	Root mean square propagation	$\theta - \text{lr} * g / \sqrt{E[g^2] + \epsilon}$	theta, gradient, mean_sq_grad, learning_rate, epsilon
725	learning_rate_decay	Learning Rate Decay	Exponential LR schedule	$\text{lr}_0 * \text{decay}^{\text{epoch}}$	initial_lr, decay_rate, epoch
726	l1_regularization	L1 Regularization (Lasso)	Absolute weight penalty	$\lambda * \text{Sum}(w)$	weights, lambda
727	l2_regularization	L2 Regularization (Ridge)	Squared weight penalty	$\lambda * \text{Sum}(w^2)$	weights, lambda
728	elastic_net_penalty	Elastic Net Penalty	L1 + L2 combined	$\lambda * (\alpha * L1 + (1 - \alpha) * L2)$	weights, lambda, alpha
729	tf_idf	TF-IDF	Term frequency inverse doc freq	$TF * \log(N / DF)$	term_freq, num_docs, doc_freq
730	cosine_sim_vectors	Cosine Similarity (Vectors)	Vector angle cosine	$A \cdot B / (\ A\ * \ B\)$	vector_a, vector_b
731	levenshtein_distance	Levenshtein Distance	Edit distance between strings	Min edits to transform	string_a, string_b
732	bleu_score	BLEU Score	Translation quality n-gram	$BP * \exp(\text{Sum}(w_n * \log(p_n)))$	reference, candidate, max_n
733	perplexity	Perplexity	Language model uncertainty	$2^{(-\text{Mean}(\log_2(p)))}$	probabilities
734	bm25_score	BM25 Score	Probabilistic relevance ranking	$IDF * (tf * (k + 1)) / (tf + k * (1 + b * dl / \text{avgdl}))$	term_freq, doc_len, avg_doc_len, idf, k, b
735	min_max_scaling	Min-Max Scaling	Scale feature to [0,1]	$(x - \min) / (\max - \min)$	x, min_val, max_val
736	standard_scaling	Standard Scaling (Z)	Zero-mean unit-variance scaling	$(x - \text{mean}) / \text{std}$	x, mean, std
737	robust_scaling	Robust Scaling	Median/IQR scaling	$(x - \text{median}) / IQR$	x, median, iqr
738	pca_explained_variance	PCA Explained Variance Ratio	Eigenvalue share	$\text{Eigenvalue}_i / \text{Sum}(\text{Eigenvalues})$	eigenvalue, total_eigenvalue_sum
739	sigmoid_derivative	Sigmoid Derivative	Gradient of sigmoid	$\text{sigmoid}(x) * (1 - \text{sigmoid}(x))$	x
740	dropout_inverted	Inverted Dropout Scale	Scale activations at train time	$x / (1 - \text{drop_rate})$	x, drop_rate
741	batch_normalization	Batch Normalization	Normalize then scale/shift	$\gamma * (x - \text{mean}) / \sqrt{\text{var} + \epsilon} + \beta$	x, mean, var, gamma, beta, eps

#	ID	Name	Description	Formula	Inputs
742	layer_normalization	Layer Normalization	Per-sample feature normalization	$\frac{\text{gamma} * (\text{x} - \text{mean})}{\sqrt{\text{var} + \text{eps}}} + \text{beta}$	x, mean, var, gamma, beta, eps
743	cosine_annealing	Cosine Annealing LR	Cosine learning-rate schedule	$\text{lr_min} + 0.5 * (\text{lr_max} - \text{lr_min}) * (1 + \cos(\pi * t / T))$	lr_min, lr_max, t, total_steps
744	attention_score	Scaled Dot-Product Attention	Softmax(QK'/sqrt(d))V weight	$\text{softmax}(QK^T / \sqrt{d_k})$	query_key_dot, d_k
745	nadam_update	NAdam Update Step	Nesterov-accelerated Adam	$\text{theta} - \text{lr} * \text{mhat} / (\sqrt{\text{vhat}} + \text{eps})$	theta, lr, mhat, vhat, eps
746	adagrad_update	AdaGrad Update Step	Per-parameter adaptive LR	$\text{theta} - \text{lr} * \text{g} / (\sqrt{\text{G}} + \text{eps})$	theta, lr, gradient, accumulated_sq, eps
747	weight_init_xavier	Xavier Init Variance	Glorot initialization variance	$2 / (\text{fan_in} + \text{fan_out})$	fan_in, fan_out
748	weight_init_he	He Init Variance	He initialization variance	$2 / \text{fan_in}$	fan_in
749	top_k_accuracy	Top-K Accuracy	Correct if true label in top k	$\text{Correct_in_TopK} / \text{Total}$	correct_in_topk, total
750	ndcg	NDCG	Normalized discounted cumulative gain	DCG / IDCG	dcg, idcg
751	map_at_k	Mean Average Precision @K	Mean of average precisions	$\text{Mean}(\text{AP@k per query})$	average_precisions
752	hamming_loss	Hamming Loss	Multi-label misclassification rate	$\text{Wrong_Labels} / \text{Total_Labels}$	wrong_labels, total_labels
753	wasserstein_distance_1d	Wasserstein Distance (1D)	Earth movers distance	$\text{Sum}(\text{CDF1} - \text{CDF2})$	cdf1, cdf2

5.15 Probability Theory & Distributions · probability.py · 55 formulas

#	ID	Name	Description	Formula	Inputs
754	normal_pdf	Normal PDF	Gaussian density	$\frac{1}{(\text{sig} * \sqrt{2\pi})} * e^{-(x - \mu)^2 / (2\text{sig}^2)}$	x, mean, std
755	normal_cdf	Normal CDF	Cumulative Gaussian	$0.5 * (1 + \text{erf}((x - \mu) / (\text{sig} * \sqrt{2})))$	x, mean, std
756	standard_normal_pdf	Standard Normal PDF	Z-distribution density	$\frac{1}{\sqrt{2\pi}} * e^{-z^2/2}$	z
757	binomial_pmf	Binomial PMF	k successes in n trials	$C(n, k) * p^k * (1 - p)^{(n - k)}$	k, n, p
758	poisson_pmf	Poisson PMF	Rare event count	$\frac{\text{lambda}^k * e^{-\text{lambda}}}{k!}$	k, lambda
759	exponential_pdf	Exponential PDF	Time between events	$\text{lambda} * e^{-\text{lambda} * x}$	x, lambda
760	uniform_pdf	Uniform PDF	Equal density over range	$\frac{1}{(b - a)}$	a, b
761	bernoulli_pmf	Bernoulli PMF	Single trial success	$p^k * (1 - p)^{(1 - k)}$	k, p
762	geometric_pmf	Geometric PMF	Trials until first success	$(1 - p)^{(k - 1)} * p$	k, p
763	negative_binomial_pmf	Negative Binomial PMF	Failures before r successes	$C(k + r - 1, k) * p^r * (1 - p)^k$	k, r, p
764	beta_pdf	Beta PDF	Bounded continuous density	$\frac{x^{a-1} * (1 - x)^{b-1}}{B(a, b)}$	x, alpha, beta
765	gamma_pdf	Gamma PDF	Waiting time density	$\frac{x^{a-1} * e^{-x/b}}{(b^a * \Gamma(a))}$	x, shape, scale
766	lognormal_pdf	Log-Normal PDF	Log-Gaussian density	$\frac{1}{(x * \text{sig} * \sqrt{2\pi})} * e^{-(\ln x - \mu)^2 / (2\text{sig}^2)}$	x, mu, sigma
767	student_t_pdf	Student's t PDF	Heavy-tail density	$\frac{\Gamma((v+1)/2)}{\Gamma(v/2)} * \frac{1}{\sqrt{v\pi}} * (1 + \frac{x^2}{v})^{-v/2}$	t, degrees_freedom
768	chi2_pdf	Chi-Square PDF	Sum of squared normals density	$\frac{x^{k/2 - 1} * e^{-x/2}}{2^{k/2} * \Gamma(k/2)}$	x, degrees_freedom
769	f_distribution_pdf	F-Distribution PDF	Variance ratio density	$\frac{\Gamma(\frac{v_1 + v_2}{2})}{\Gamma(\frac{v_1}{2}) * \Gamma(\frac{v_2}{2})} * (\frac{v_1}{v_2})^{v_1/2} * (1 + \frac{v_1}{v_2})^{-(v_1 + v_2)/2}$	x, df1, df2
770	weibull_pdf	Weibull PDF	Reliability/failure density	$(k/l) * (x/l)^{(k-1)} * e^{-(x/l)^k}$	x, shape, scale
771	conditional_probability	Conditional Probability	P(A given B)	$\frac{P(A \text{ and } B)}{P(B)}$	p_a_and_b, p_b
772	bayes_theorem	Bayes' Theorem	Posterior probability	$\frac{P(B A) * P(A)}{P(B)}$	p_b_given_a, p_a, p_b
773	joint_probability_independent	Joint Probability (Independent)	P(A and B) when independent	$P(A) * P(B)$	p_a, p_b
774	union_probability	Union Probability	P(A or B)	$P(A) + P(B) - P(A \text{ and } B)$	p_a, p_b, p_a_and_b
775	complement_probability	Complement Probability	P(not A)	$1 - P(A)$	p_a

#	ID	Name	Description	Formula	Inputs
776	total_probability	Total Probability	Marginalize over partition	$\text{Sum}(P(A \setminus B_i) * P(B_i))$	conditionals, priors
777	odds_from_probability	Odds from Probability	Convert probability to odds	$p / (1 - p)$	probability
778	probability_from_odds	Probability from Odds	Convert odds to probability	$\text{odds} / (1 + \text{odds})$	odds
779	permutations_count	Permutations Count	Ordered arrangements	$n! / (n-r)!$	n, r
780	combinations_count	Combinations Count	Unordered selections	$n! / (r!(n-r)!)$	n, r
781	multinomial_coefficient	Multinomial Coefficient	Multi-group arrangements	$n! / (n_1! * n_2! * \dots * n_k!)$	n, group_sizes
782	permutations_with_repetition	Permutations with Repetition	Arrangements with repeats	n^r	n, r
783	circular_permutations	Circular Permutations	Round-table arrangements	$(n-1)!$	n
784	expected_value_discrete	Expected Value (Discrete)	Probability-weighted mean	$\text{Sum}(x_i * p_i)$	values, probabilities
785	variance_discrete	Variance (Discrete)	Expected squared deviation	$\text{Sum}(p * (x - \mu)^2)$	values, probabilities
786	covariance_random_vars	Covariance (Random Vars)	Joint expectation deviation	$E[XY] - E[X]E[Y]$	joint_values, probabilities
787	correlation_random_vars	Correlation (Random Vars)	Normalized covariance	$\text{Cov}(X, Y) / (\text{sig}X * \text{sig}Y)$	covariance, std_x, std_y
788	moment_generating	Moment (n-th)	n-th raw moment	$E[X^n]$	values, probabilities, n
789	variance_sum_independent	Variance of Sum (Independent)	Sum of variances	$\text{Var}(X) + \text{Var}(Y)$	var_x, var_y
790	markov_steady_state	Markov Steady State	Stationary distribution	$\pi_i = \pi_i * P$	transition_matrix
791	poisson_process_prob	Poisson Process Probability	Events in interval	$(\text{lambda} * t)^k * e^{-(\text{lambda} * t)} / k!$	rate, time, k
792	geometric_brownian_motion	Geometric Brownian Motion	Stock price diffusion	$S_0 * e^{((\mu - \text{sig}^2/2) * t + \text{sig} * W)}$	s0, mu, sigma, time, wiener
793	random_walk_position	Random Walk Position	Cumulative step position	$\text{Sum}(\text{steps})$	steps
794	chebyshev_inequality	Chebyshev Inequality	Bound on tail probability	$1 / k^2$	k
795	markov_inequality	Markov Inequality	Bound $P(X \geq a)$	Mean / a	mean, a
796	central_limit_theorem	CLT Sampling Std	Std of sample mean	$\text{sigma} / \text{sqrt}(n)$	sigma, n
797	bayesian_posterior	Bayesian Posterior	Posterior via Bayes rule	$(\text{Likelihood} * \text{Prior}) / \text{Evidence}$	likelihood, prior, evidence
798	law_large_numbers_error	LLN Convergence Error	Sample mean deviation bound	$\text{sigma} / \text{sqrt}(n)$	sigma, n
799	hypergeometric_pmf	Hypergeometric PMF	Draws without replacement	$C(K, k) C(N-K, n-k) / C(N, n)$	population, successes, draws, observed
800	multinomial_pmf	Multinomial PMF	Multi-category probability	$n! / \text{Prod}(x_i!) * \text{Prod}(\pi_i^{x_i})$	n, counts, probabilities
801	cauchy_pdf	Cauchy PDF	Heavy-tailed density	$1 / (\pi * \text{gamma} * (1 + ((x - x_0) / \text{gamma})^2))$	x, location, scale
802	pareto_pdf	Pareto PDF	Power-law density	$\text{alpha} * x^{-\text{alpha}} / x^{(\text{alpha}+1)}$	x, scale_min, alpha
803	survival_function	Survival Function	Probability of exceeding t	$1 - \text{CDF}(t)$	cdf
804	hazard_rate	Hazard Rate	Instantaneous failure rate	$\text{pdf}(t) / \text{Survival}(t)$	pdf, survival
805	entropy_shannon	Shannon Entropy	Information content of distribution	$-\text{Sum}(p * \log_2(p))$	probabilities
806	cross_entropy_dist	Cross Entropy (Distributions)	Between true and predicted	$-\text{Sum}(p * \log(q))$	p_true, q_pred
807	conditional_variance	Conditional Variance	Var of X given Y	$E[X^2 \setminus Y] - E[X \setminus Y]^2$	e_x2_given_y, e_x_given_y
808	expected_shortfall_prob	Tail Expectation	Expected value beyond threshold	$E[X \setminus X > \text{threshold}]$	tail_values

5.16 Advanced Statistics & Hypothesis Testing · `statistics_advanced.py` · 53 formulas

#	ID	Name	Description	Formula	Inputs
809	<code>t_test_one_sample</code>	One-Sample t-Test	Mean vs hypothesized value	$(\bar{x} - \mu) / (s/\sqrt{n})$	sample_mean, pop_mean, std, n
810	<code>t_test_two_sample</code>	Two-Sample t-Test	Compare two means	$(x_1 - x_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$	mean1, mean2, std1, std2, n1, n2
811	<code>paired_t_test</code>	Paired t-Test	Compare paired observations	$\bar{d} / (sd/\sqrt{n})$	mean_diff, std_diff, n
812	<code>welch_t_test</code>	Welch's t-Test	Unequal-variance t-test	$(x_1 - x_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$	mean1, mean2, var1, var2, n1, n2
813	<code>z_test_proportion</code>	Z-Test for Proportion	Sample proportion test	$(\hat{p} - p_0) / \sqrt{p_0(1-p_0)/n}$	sample_prop, pop_prop, n
814	<code>z_test_mean</code>	Z-Test for Mean	Known-variance mean test	$(\bar{x} - \mu) / (\sigma/\sqrt{n})$	sample_mean, pop_mean, sigma, n
815	<code>anova_f_statistic</code>	ANOVA F-Statistic	Between vs within variance	MSB / MSW	between_group_var, within_group_var
816	<code>chi2_independence</code>	Chi-Square Independence	Test variable association	$\sum ((O-E)^2/E)$	observed, expected
817	<code>chi2_goodness_of_fit</code>	Chi-Square Goodness of Fit	Observed vs expected	$\sum ((O-E)^2/E)$	observed, expected
818	<code>mann_whitney_u</code>	Mann-Whitney U	Nonparametric rank test	$U = R_1 - n_1(n_1+1)/2$	ranks, n1, n2
819	<code>wilcoxon_signed_rank</code>	Wilcoxon Signed-Rank	Paired nonparametric test	Sum of signed ranks	differences
820	<code>kruskal_wallis_h</code>	Kruskal-Wallis H	Multi-group nonparametric	$12/(N(N+1)) * \sum (R_i^2/n_i) - 3(N+1)$	rank_sums, group_sizes, n_total
821	<code>levene_test</code>	Levene's Test	Equality of variances	F-stat on abs deviations	groups
822	<code>f_test_variance</code>	F-Test for Variances	Ratio of two variances	s_1^2 / s_2^2	variance1, variance2
823	<code>cohens_d</code>	Cohen's d	Standardized mean difference	$(\text{mean1} - \text{mean2}) / \text{pooled_std}$	mean1, mean2, pooled_std
824	<code>hedges_g</code>	Hedges' g	Bias-corrected Cohen's d	$\text{Cohens_d} * (1 - 3/(4df-1))$	cohens_d, degrees_freedom
825	<code>eta_squared</code>	Eta Squared	ANOVA effect size	$SS_{\text{between}} / SS_{\text{total}}$	ss_between, ss_total
826	<code>odds_ratio</code>	Odds Ratio	Exposure-outcome association	$(a*d) / (b*c)$	a, b, c, d
827	<code>relative_risk</code>	Relative Risk	Risk ratio between groups	$(a/(a+b)) / (c/(c+d))$	a, b, c, d
828	<code>confidence_interval_mean</code>	CI for Mean	Mean confidence bounds	$\bar{x} \pm t*(s/\sqrt{n})$	mean, std, n, confidence
829	<code>confidence_interval_proportion</code>	CI for Proportion	Proportion confidence bounds	$\hat{p} \pm z*\sqrt{\hat{p}(1-\hat{p})/n}$	proportion, n, confidence
830	<code>margin_of_error</code>	Margin of Error	Half-width of CI	$z * (std / \sqrt{n})$	z_score, std, n
831	<code>prediction_interval</code>	Prediction Interval	Future observation bounds	$\hat{y} \pm t*s*\sqrt{1+1/n+\dots}$	prediction, std_error, n, confidence
832	<code>p_value_from_z</code>	P-Value from Z	Tail probability from z	$2*(1 - \Phi(z))$	z_score
833	<code>logistic_regression_prob</code>	Logistic Regression Probability	Sigmoid of linear combo	$1/(1+e^{-(b_0+b_1*x)})$	intercept, coefficient, x
834	<code>multiple_regression_predict</code>	Multiple Regression Prediction	Linear combination of features	$b_0 + \sum(b_i*x_i)$	intercept, coefficients, features
835	<code>ridge_penalty_cost</code>	Ridge Regression Cost	SSE plus L2 penalty	$SSE + \lambda \sum(b_i^2)$	sse, coefficients, lambda
836	<code>vif</code>	Variance Inflation Factor	Multicollinearity measure	$1 / (1 - R^2_i)$	r_squared_i
837	<code>partial_correlation</code>	Partial Correlation	Correlation controlling for Z	$(r_{xy} - r_{xz}*r_{yz})/\sqrt{\dots}$	rx_y, rx_z, ry_z
838	<code>durbin_watson_test</code>	Durbin-Watson	Residual autocorrelation	$\sum((e_t - e_{t-1})^2) / \sum(e_t^2)$	residuals
839	<code>standardized_residual</code>	Standardized Residual	Residual / std error	$\text{residual} / \text{std_error}$	residual, std_error
840	<code>leverage_hat</code>	Leverage (Hat Value)	Observation influence	Diagonal of hat matrix	x_matrix, observation
841	<code>cooks_distance</code>	Cook's Distance	Influence of observation	$(e^2/(p*MSE))*(h/(1-h)^2)$	residual, leverage, p, mse
842	<code>sample_size_mean</code>	Sample Size for Mean	n for desired margin	$(z*\sigma/E)^2$	z_score, std, margin_error

#	ID	Name	Description	Formula	Inputs
843	sample_size_proportion	Sample Size for Proportion	n for proportion estimate	$z^2 * p(1-p) / E^2$	z_score, proportion, margin_error
844	standard_error_proportion	Standard Error of Proportion	Proportion sampling error	$\sqrt{p(1-p)/n}$	proportion, n
845	finite_population_correction	Finite Population Correction	FPC factor	$\sqrt{(N-n)/(N-1)}$	population, sample
846	bootstrap_std_error	Bootstrap Standard Error	Resampling-based SE	Std of bootstrap statistics	bootstrap_estimates
847	pooled_variance	Pooled Variance	Combined sample variance	$((n1-1)s1^2 + (n2-1)s2^2) / (n1+n2-2)$	var1, var2, n1, n2
848	spearman_rank	Spearman Rank Correlation	Monotonic correlation	$1 - 6 * \text{Sum}(d^2) / (n(n^2-1))$	rank_diffs, n
849	kendall_tau_b	Kendall Tau-b	Ordinal association with ties	$(C - D) / \sqrt{(C+D+T)(C+D+U)}$	concordant, discordant, ties_x, ties_y
850	point_biserial	Point-Biserial Correlation	Binary vs continuous correlation	$(M1-M0) / \text{Std} * \sqrt{p*q}$	mean1, mean0, std, p, q
851	shapiro_wilk_stat	Shapiro-Wilk Statistic	Normality test W	$(\text{Sum}(a_i * x_i))^2 / \text{Sum}((x - \bar{x})^2)$	ordered_values, coefficients
852	kolmogorov_smirnov	Kolmogorov-Smirnov D	Max CDF deviation	$\max F_{\text{empirical}} - F_{\text{theoretical}} $	empirical_cdf, theoretical_cdf
853	jarque_bera	Jarque-Bera Statistic	Normality from skew/kurtosis	$n/6 * (S^2 + (K-3)^2/4)$	n, skewness, kurtosis
854	bonferroni_correction	Bonferroni Alpha	Adjusted significance level	α / m	alpha, num_tests
855	benjamini_hochberg	Benjamini-Hochberg Threshold	FDR adjusted critical value	$(i/m) * \alpha$	rank, num_tests, alpha
856	tukey_hsd	Tukey HSD Critical Diff	Honest significant difference	$q * \sqrt{\text{MSE}/n}$	q_critical, mse, n
857	power_analysis	Statistical Power	1 minus beta	$1 - \text{Beta}$	beta
858	kaplan_meier	Kaplan-Meier Survival	Product-limit survival estimate	$\text{Prod}((n_i - d_i) / n_i)$	at_risk, events
859	gini_coefficient_stat	Gini Coefficient	Inequality measure 0-1	Sum of Lorenz deviations	values
860	theil_index	Theil Index	Entropy-based inequality	$\text{Mean}((x/\bar{x}) * \ln(x/\bar{x}))$	values
861	cohens_kappa_stat	Cohen's Kappa	Inter-rater agreement	$(Po - Pe) / (1 - Pe)$	observed_agreement, expected_agreement

5.17 Business Analyst KPIs, Forecasting & Decision · business_analyst.py · 49 formulas

#	ID	Name	Description	Formula	Inputs
862	conversion_rate	Conversion Rate %	Conversions / total visitors	$\text{Conversions} / \text{Visitors} * 100$	conversions, visitors
863	retention_rate	Retention Rate %	Retained / starting customers	$(\text{End} - \text{New}) / \text{Start} * 100$	start_customers, end_customers, new_customers
864	churn_rate	Churn Rate %	Lost / starting customers	$\text{Lost_Customers} / \text{Start_Customers} * 100$	lost_customers, start_customers
865	customer_lifetime_value	Customer Lifetime Value	Total value per customer	$\text{ARPU} * \text{Gross_Margin} / \text{Churn_Rate}$	arpu, gross_margin, churn_rate
866	cac	Customer Acquisition Cost	Sales+marketing / new customers	$\text{Total_Sales_Marketing} / \text{New_Customers}$	sales_marketing_cost, new_customers
867	cac_payback_period	CAC Payback Period (months)	Months to recover CAC	$\text{CAC} / (\text{ARPU} * \text{Gross_Margin})$	cac, arpu, gross_margin
868	net_promoter_score	Net Promoter Score	Promoters minus detractors %	$(\text{Promoters} - \text{Detractors}) / \text{Total} * 100$	promoters, detractors, total
869	market_share	Market Share %	Company sales / market sales	$\text{Company_Sales} / \text{Market_Sales} * 100$	company_sales, market_sales
870	wallet_share	Share of Wallet %	Our spend / total customer spend	$\text{Customer_Spend_With_Us} / \text{Total_Customer_Spend} * 100$	spend_with_us, total_spend
871	funnel_conversion	Funnel Conversion %	Stage-to-stage conversion	$\text{Stage_N} / \text{Stage_1} * 100$	stage_n, stage_1

#	ID	Name	Description	Formula	Inputs
872	active_user_ratio	Active User Ratio (DAU/MAU)	Stickiness measure	DAU / MAU	dau, mau
873	engagement_rate	Engagement Rate %	Engaged / total users	$Engaged_Users / Total_Users * 100$	engaged_users, total_users
874	bounce_rate	Bounce Rate %	Single-page sessions / total	$Single_Page_Sessions / Total_Sessions * 100$	single_page_sessions, total_sessions
875	cohort_retention	Cohort Retention %	Active cohort / original cohort	$Active_In_Period / Original_Cohort * 100$	active_in_period, original_cohort
876	linear_forecast	Linear Forecast	Trend-line projection	$intercept + slope * period$	intercept, slope, period
877	seasonal_index	Seasonal Index	Seasonal adjustment factor	$Period_Average / Overall_Average$	period_average, overall_average
878	weighted_moving_forecast	Weighted Moving Forecast	Weighted recent values	$Sum(weight*value) / Sum(weights)$	values, weights
879	forecast_bias	Forecast Bias	Average forecast error	$Sum(Actual - Forecast) / n$	actuals, forecasts
880	tracking_signal	Tracking Signal	Cumulative bias / MAD	$Cumulative_Error / MAD$	cumulative_error, mad
881	mean_absolute_deviation	Mean Absolute Deviation	Average absolute forecast error	$Mean(Actual - Forecast)$	actuals, forecasts
882	exponential_smoothing_forecast	Exponential Smoothing Forecast	Weighted recency forecast	$alpha*actual + (1-alpha)*prev_forecast$	actual, prev_forecast, alpha
883	expected_monetary_value	Expected Monetary Value	Probability-weighted payoff	$Sum(Probability * Payoff)$	probabilities, payoffs
884	value_of_information	Expected Value of Information	Value of perfect info	$EV_With_Info - EV_Without_Info$	ev_with_info, ev_without_info
885	decision_tree_value	Decision Tree Node Value	Best expected branch value	$Max(branch\ EMVs)$	branch_values
886	regret_value	Maximum Regret	Opportunity loss measure	$Max(Best_Payoff - Chosen_Payoff)$	payoff_matrix, chosen
887	sensitivity_elasticity	Sensitivity Elasticity	% output change / % input change	$Pct_Change_Output / Pct_Change_Input$	pct_change_output, pct_change_input
888	breakeven_units_ba	Breakeven Units	Fixed costs / unit margin	$Fixed_Costs / (Price - Variable_Cost)$	fixed_costs, price, variable_cost
889	roi_business	Return on Investment %	Net gain / cost	$(Gain - Cost) / Cost * 100$	gain, cost
890	tam_sam_som	TAM/SAM/SOM	Serviceable obtainable market	$TAM * SAM_Pct * SOM_Pct$	tam, sam_pct, som_pct
891	price_elasticity_demand	Price Elasticity of Demand	% qty change / % price change	$Pct_Change_Qty / Pct_Change_Price$	pct_change_qty, pct_change_price
892	cross_price_elasticity	Cross-Price Elasticity	Demand sensitivity to other price	$Pct_Change_Qty_A / Pct_Change_Price_B$	pct_change_qty_a, pct_change_price_b
893	income_elasticity	Income Elasticity	Demand sensitivity to income	$Pct_Change_Qty / Pct_Change_Income$	pct_change_qty, pct_change_income
894	economic_order_quantity	Economic Order Quantity	Optimal order size	$sqrt(2*D*S / H)$	annual_demand, order_cost, holding_cost
895	reorder_point	Reorder Point	Inventory trigger level	$Daily_Demand * Lead_Time + Safety_Stock$	daily_demand, lead_time, safety_stock
896	safety_stock	Safety Stock	Buffer inventory	$Z * sigma * sqrt(Lead_Time)$	z_service, demand_std, lead_time
897	capacity_utilization	Capacity Utilization %	Actual vs potential output	$Actual_Output / Potential_Output * 100$	actual_output, potential_output
898	learning_curve	Learning Curve Unit Cost	Cost decline with volume	$First_Cost * Units^{(log(rate)/log(2))}$	first_unit_cost, cumulative_units, learning_rate
899	gmV	Gross Merchandise Value	Total marketplace sales	$Sum(Order_Values)$	order_values
900	take_rate	Take Rate %	Platform revenue / GMV	$Revenue / GMV * 100$	revenue, gmV
901	average_order_value	Average Order Value	Revenue per order	$Total_Revenue / Order_Count$	total_revenue, order_count
902	repeat_purchase_rate	Repeat Purchase Rate %	Returning customers share	$Repeat_Customers / Total_Customers * 100$	repeat_customers, total_customers

#	ID	Name	Description	Formula	Inputs
903	attribution_linear	Linear Attribution Credit	Equal credit per touchpoint	$\text{Conversion_Value} / \text{Touchpoints}$	conversion_value, touchpoints
904	roi_marketing	Marketing ROI %	Profit from marketing / spend	$(\text{Revenue} - \text{Cost}) / \text{Cost} * 100$	revenue, cost
905	roas	Return on Ad Spend	Revenue per ad dollar	$\text{Ad_Revenue} / \text{Ad_Spend}$	ad_revenue, ad_spend
906	ltv_cac_payback	LTV/CAC Payback (months)	Months to recover CAC	$\text{CAC} / (\text{ARPU} * \text{Gross_Margin})$	cac, arpu, gross_margin
907	cash_runway_months	Cash Runway (months)	Months until cash out	$\text{Cash} / \text{Monthly_Burn}$	cash, monthly_burn
908	weighted_pipeline	Weighted Sales Pipeline	Probability-weighted deals	$\text{Sum}(\text{Deal_Value} * \text{Win_Probability})$	deal_values, win_probabilities
909	win_rate	Sales Win Rate %	Won deals / total deals	$\text{Won_Deals} / \text{Total_Deals} * 100$	won_deals, total_deals
910	market_growth_rate	Market Growth Rate %	YoY market size change	$(\text{Market_Now} - \text{Market_Prior}) / \text{Market_Prior} * 100$	market_now, market_prior

6. Testing Strategy

Testing is **mandatory** and gates every formula. No formula is considered "done" until its tests pass.

6.1 Test Requirements Per Formula

Each of the 610 formulas must have **at least 3 test cases**:

1. **Known-answer (golden) test** — a textbook/real-filing example where the correct answer is independently known. Example: Apple FY2023 gross margin = 44.13% from the 10-K.
2. **Edge case** — zero denominators, negative values, empty series, single-element series. Must return `valid=False` gracefully, never crash.
3. **Property/invariant test** — a mathematical property that must hold. Example: `gross_margin <= 100`, `current_ratio > 0` for positive inputs, `put_call_parity` residual ≈ 0 .

Minimum test count: $910 \times 3 = 2730$ test cases.

6.2 Test File Pattern

```
import pytest
from src.maths_lib.profitability import gross_margin

def test_gross_margin_known_answer():
    # Apple FY2023: Revenue 383,285 COGS 214,137
    r = gross_margin(revenue=383285, cogs=214137)
    assert r.valid is True
    assert abs(r.value - 44.13) < 0.01      # tolerance 0.01pp
    assert r.unit == '%'

def test_gross_margin_zero_revenue():
    r = gross_margin(revenue=0, cogs=100)
    assert r.valid is False                # division guard
    assert r.value is None

def test_gross_margin_invariant():
    r = gross_margin(revenue=1000, cogs=300)
    assert r.value <= 100                  # margin can't exceed 100%
```

6.3 Numerical Tolerance Standards

Formula Type	Tolerance	Reason
Ratios, margins (%)	± 0.01	Rounding in source filings
Currency values (\$)	± 0.5 or $\pm 0.01\%$	Rounding to nearest unit
Option prices (Black-Scholes)	± 0.001	Sensitive to vol/time
Monte Carlo / VaR	$\pm 2\%$ (seed=42)	Stochastic, fixed-seed
Duration / convexity	± 0.001	Bond math precision
Trig / pure math	$\pm 1e-9$	Floating-point limit

6.4 Golden Reference Set

`tests/maths_lib/golden/` holds JSON files with known-answer inputs and outputs sourced from: real SEC 10-K filings (Apple, NVIDIA, 3M), CFA curriculum examples, and standard textbook problems (Hull for options, Fabozzi for fixed income). These are the regression anchors.

6.5 CI Gate

```
# Run the full maths_lib test suite
pytest tests/maths_lib/ -v --tb=short

# Coverage requirement: 100% line coverage on maths_lib/
pytest tests/maths_lib/ --cov=src.maths_lib --cov-fail-under=100

# Registry integrity (no duplicate IDs, every fn registered)
pytest tests/maths_lib/test_registry.py -v
```

The build does not advance to the next domain until the current domain's tests are green.

7. Registry & Lookup API (`registry.py`)

The registry is how the Composite Resolver (N20) and Formula Router find formulas at runtime.

```
from src.maths_lib.registry import (
    get_formula, search_by_keyword, list_by_domain, compute
)

# Direct lookup by ID
fn = get_formula('black_scholes_call')
result = fn(spot=100, strike=105, time=0.5, rate=0.03, volatility=0.2)

# Keyword search (for Composite Resolver side-word matching)
matches = search_by_keyword('turnover')
# -> ['asset_turnover', 'inventory_turnover', 'receivables_turnover', ...]

# Compute by ID with a dict of inputs (used by N20 matrix solver)
result = compute('current_ratio',
    {'current_assets': 143566, 'current_liabilities': 145308})
# -> FormulaResult(value=0.988, unit='x', ...)

# List a whole domain
ratio_fns = list_by_domain('D01_profitability')
```

Registry guarantees:

- Every formula ID is globally unique (enforced by `test_registry.py`).
- Every registered function has metadata: name, expression, domain, unit, input list.
- `compute(fid, inputs)` validates that all required inputs are present before calling.

8. Integration With the FinBench Pipeline

`maths_lib` plugs into two pipeline nodes:

8.1 Formula Router (between N06 Sniper and N07 BM25)

When a question contains formula keywords (`ratio`, `margin`, `turnover`, `duration`, `Black-Scholes`, etc.), the Formula Router:

1. Identifies the target formula via `search_by_keyword`.
2. Uses Sniper to extract the required input values from the filing.
3. Calls `compute(fid, inputs)`.
4. Returns a deterministic `FormulaResult` — bypassing the LLM entirely.

8.2 Composite Resolver / N20 (the JEE side-word method)

For multi-step questions ("5-in-1" formulas), N20:

1. Harvests side-words to detect which sub-formulas are needed.
2. Builds a dependency graph among `maths_lib` functions.
3. Anchors on the single given value (Sniper-extracted).
4. Topologically sorts and calls sub-formulas in order.

5. Self-verifies via 3 independent computation paths.

Because every `maths_lib` function returns inputs + expression, N20 can emit a full audit trail — a key differentiator for enterprise/audit use.

9. Build Plan (Session-by-Session)

One domain per session. Each session: write module -> write tests -> run tests green -> commit -> update CONTEXT.md.

Session	Domain	Module	Formulas	Cumulative
1	Profitability & Margin Ratios	<code>profitability.py</code>	50	50
2	Liquidity, Solvency & Efficiency Ratios	<code>liquidity_solvency.py</code>	50	100
3	Valuation Metrics & Models	<code>valuation.py</code>	60	160
4	Time-Series & Technical Analysis	<code>technical.py</code>	80	240
5	Options Pricing & Derivatives	<code>options.py</code>	45	285
6	Fixed Income & Bonds	<code>fixed_income.py</code>	50	335
7	Risk Management & Portfolio Theory	<code>risk.py</code>	50	385
8	Time Value of Money & Capital Budgeting	<code>tvm.py</code>	45	430
9	Corporate Finance & M&A	<code>corporate_ma.py</code>	50	480
10	Accounting & Depreciation	<code>accounting.py</code>	40	520
11	Statistics & Econometrics	<code>statistics_econ.py</code>	45	565
12	Core Math, Trigonometry, Linear Algebra & Geometry	<code>math_core.py</code>	45	610
13	Growth, Segment, Forensic & Modern Metrics	<code>growth_segment_forensic.py</code>	50	660
14	AI / Machine Learning Metrics & Functions	<code>ai_ml.py</code>	93	753
15	Probability Theory & Distributions	<code>probability.py</code>	55	808
16	Advanced Statistics & Hypothesis Testing	<code>statistics_advanced.py</code>	53	861
17	Business Analyst KPIs, Forecasting & Decision	<code>business_analyst.py</code>	49	910

At your stated pace (~30 formulas/day), the remaining work is roughly **910 total**, and you reported ~150 already done — so about **760 remaining ≈ 15 days** of focused building, including tests.

10. Constraints Compliance

Constraint	How <code>maths_lib</code> complies
C1 Zero Cost	Pure Python/NumPy/SciPy — no paid APIs
C2 Local Inference	No network calls; all computation in-process
C5 Deterministic Seed	<code>seed=42</code> for Monte Carlo, simulations
C9 Private Fields	<code>maths_lib</code> never reads/writes <code>_rlef_</code> fields
A-perf	Each formula < 10 ms execution

11. Acceptance Criteria (Definition of Done)

- [] All 910 formulas implemented across 12 modules.
- [] All 910 formulas registered in `FORMULA_REGISTRY` with unique IDs.
- [] ≥ 2730 test cases, all passing.
- [] 100% line coverage on `src/maths_lib/`.
- [] Golden reference set populated from real filings.
- [] Registry integrity test passing (unique IDs, all functions registered).
- [] Formula Router integrated and tested on FinanceBench ratio questions.
- [] Every `FormulaResult` carries name + expression + inputs (audit trail).
- [] No formula exceeds 10 ms execution time.

End of PDR-MATHSLIB-001 Rev 1.0 — 910 formulas specified.